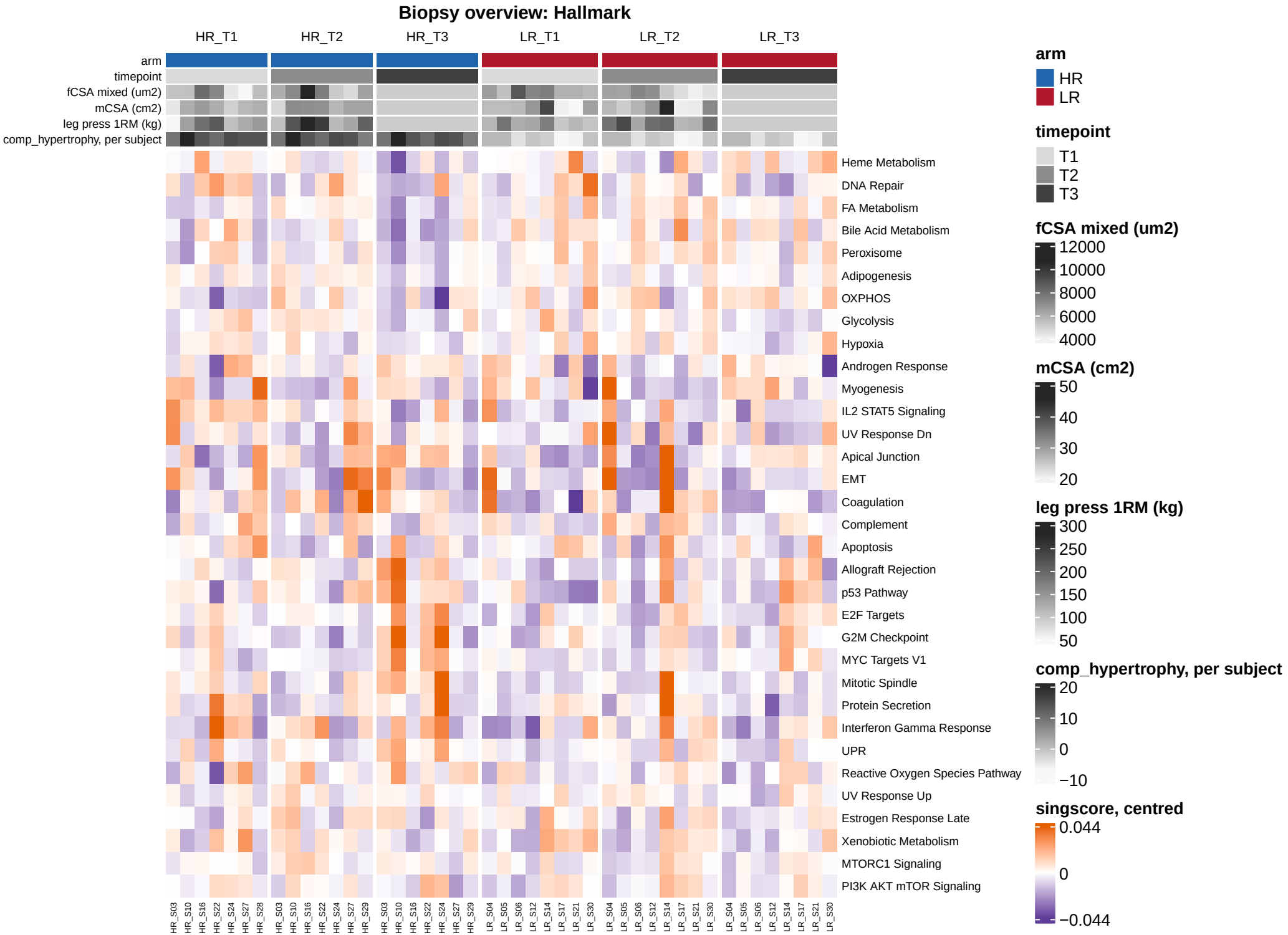
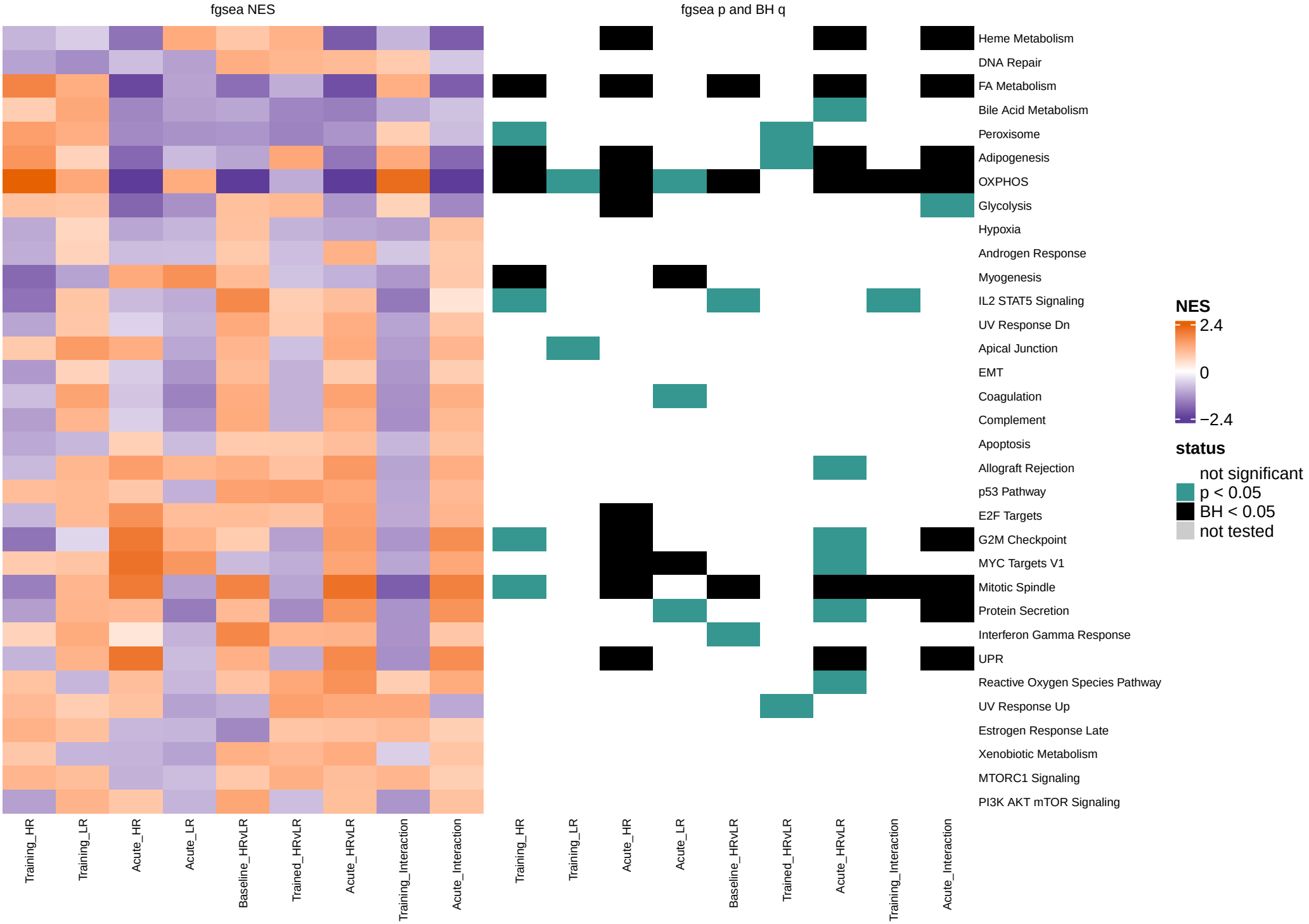
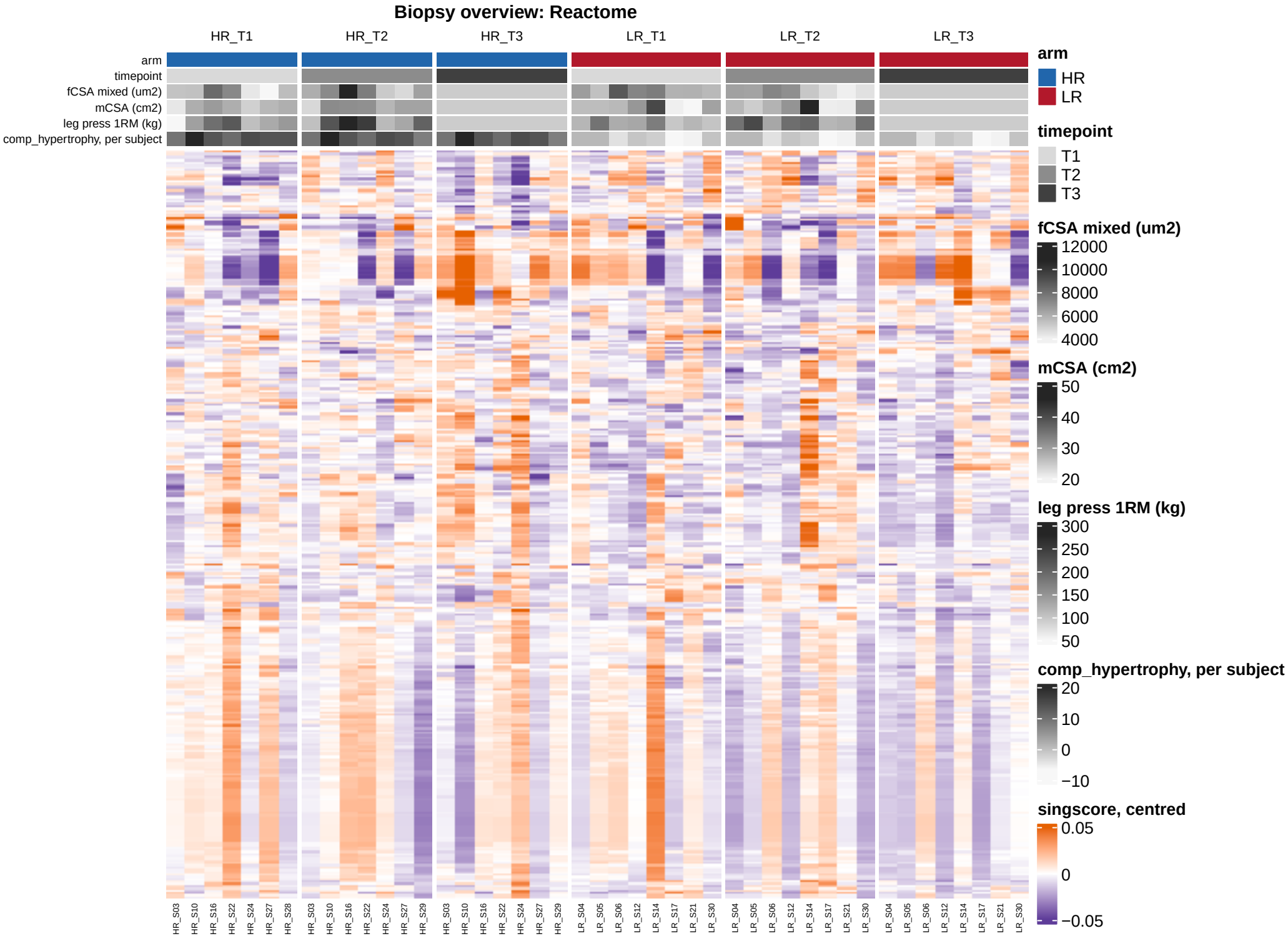


All 33 sets by 45 biopsies, grouped by arm and time point; column labels are subjects. Fill: singscore minus the set's mean over biopsies. Colours stop at the 99th percentile of |value|, 0.044. Rows are ordered within collection by average−linkage clustering (1 − Pearson r) of singscore, for display only. Grey: missing.

Contrast overview: Hallmark

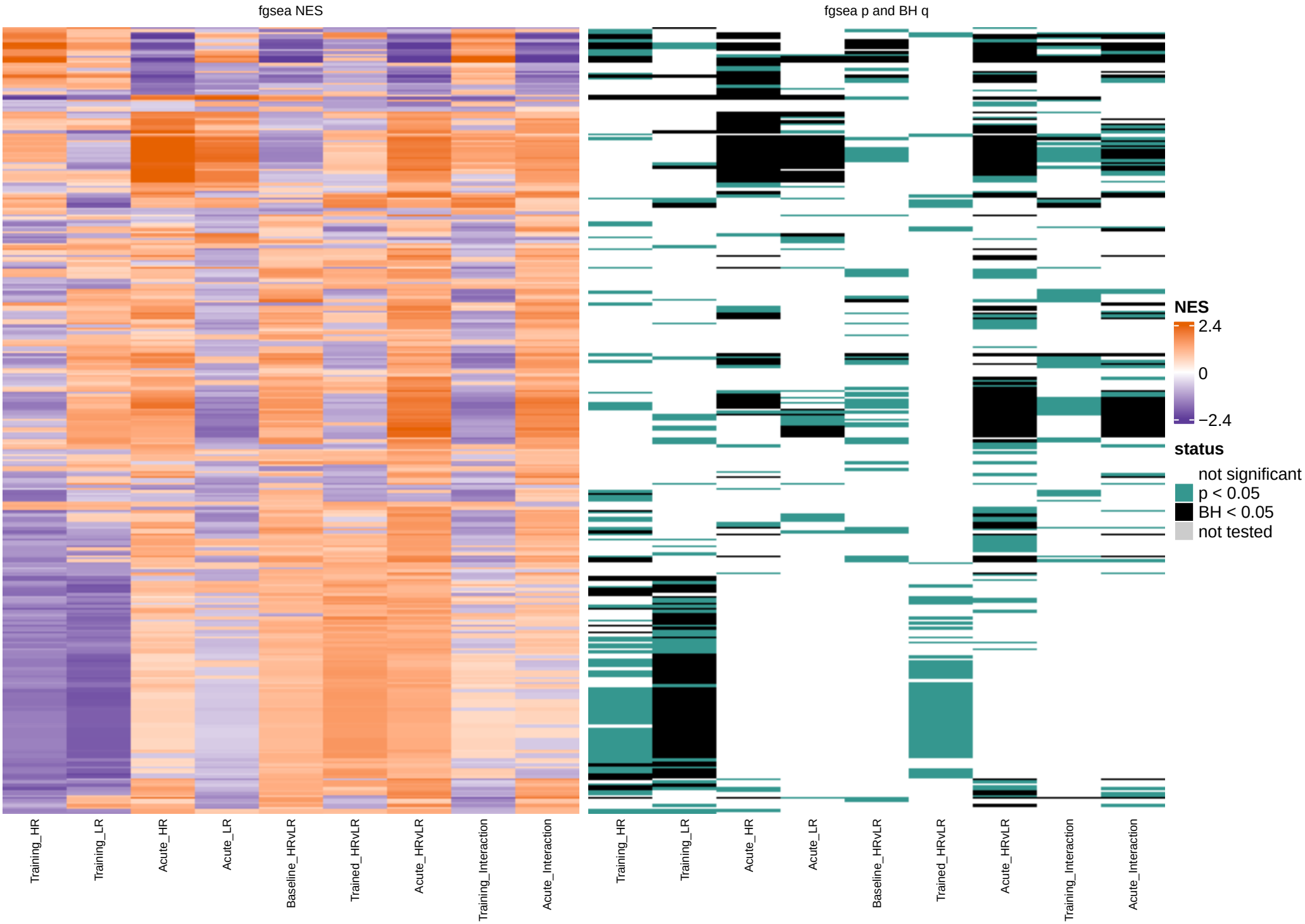


All 33 sets by the nine contrasts. Fill: fgsea normalised enrichment score; status: p and BH q within contrast, collections pooled. Colours stop at the 99th percentile of |value|, 2.4. Rows are ordered within collection by average−linkage clustering (1 − Pearson r) of singscore, for display only. Grey: not tested. Table: c_data/set_tests.rds, set_tests.

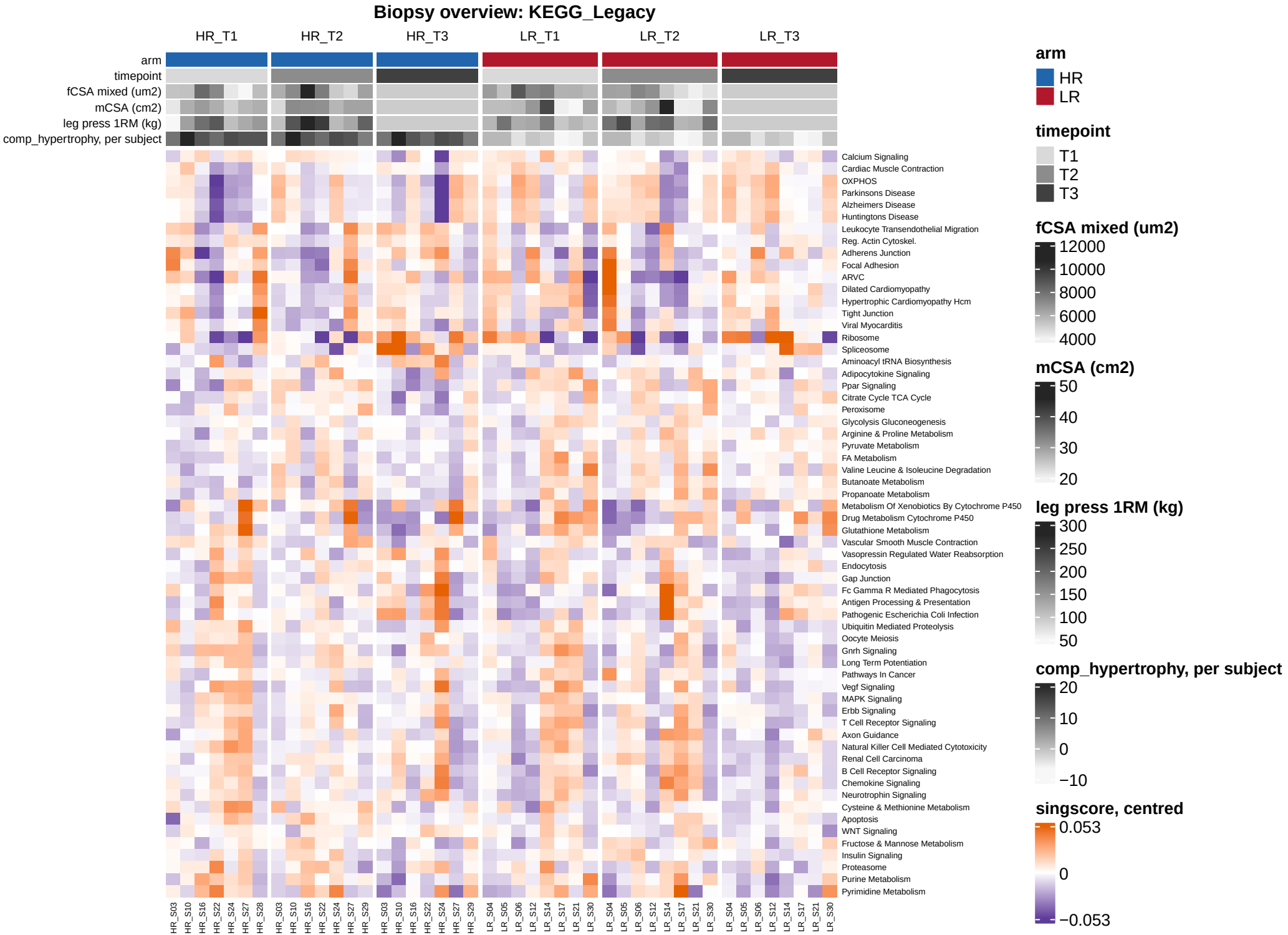


All 328 sets by 45 biopsies, grouped by arm and time point; column labels are subjects. Fill: singscore minus the set's mean over biopsies. Colours stop at the 99th percentile of |value|, 0.05. Rows are ordered within collection by average−linkage clustering (1 − Pearson r) of singscore, for display only. Grey: missing.

Contrast overview: Reactome

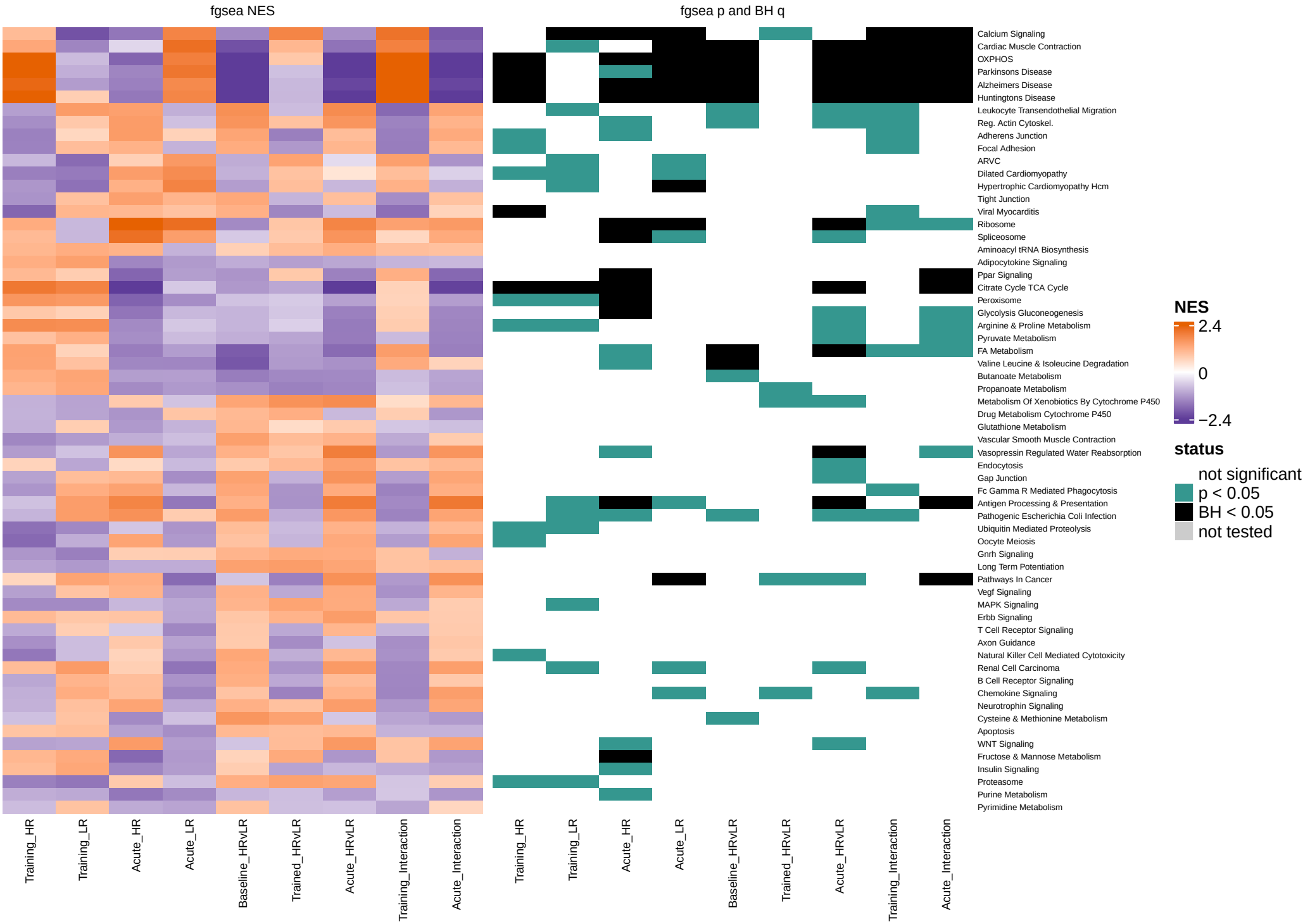


All 328 sets by the nine contrasts. Fill: fgsea normalised enrichment score; status: p and BH q within contrast, collections pooled. Colours stop at the 99th percentile of |value|, 2.4. Rows are ordered within collection by average−linkage clustering (1 − Pearson r) of singscore, for display only. Grey: not tested. Table: c_data/set_tests.rds, set_tests.

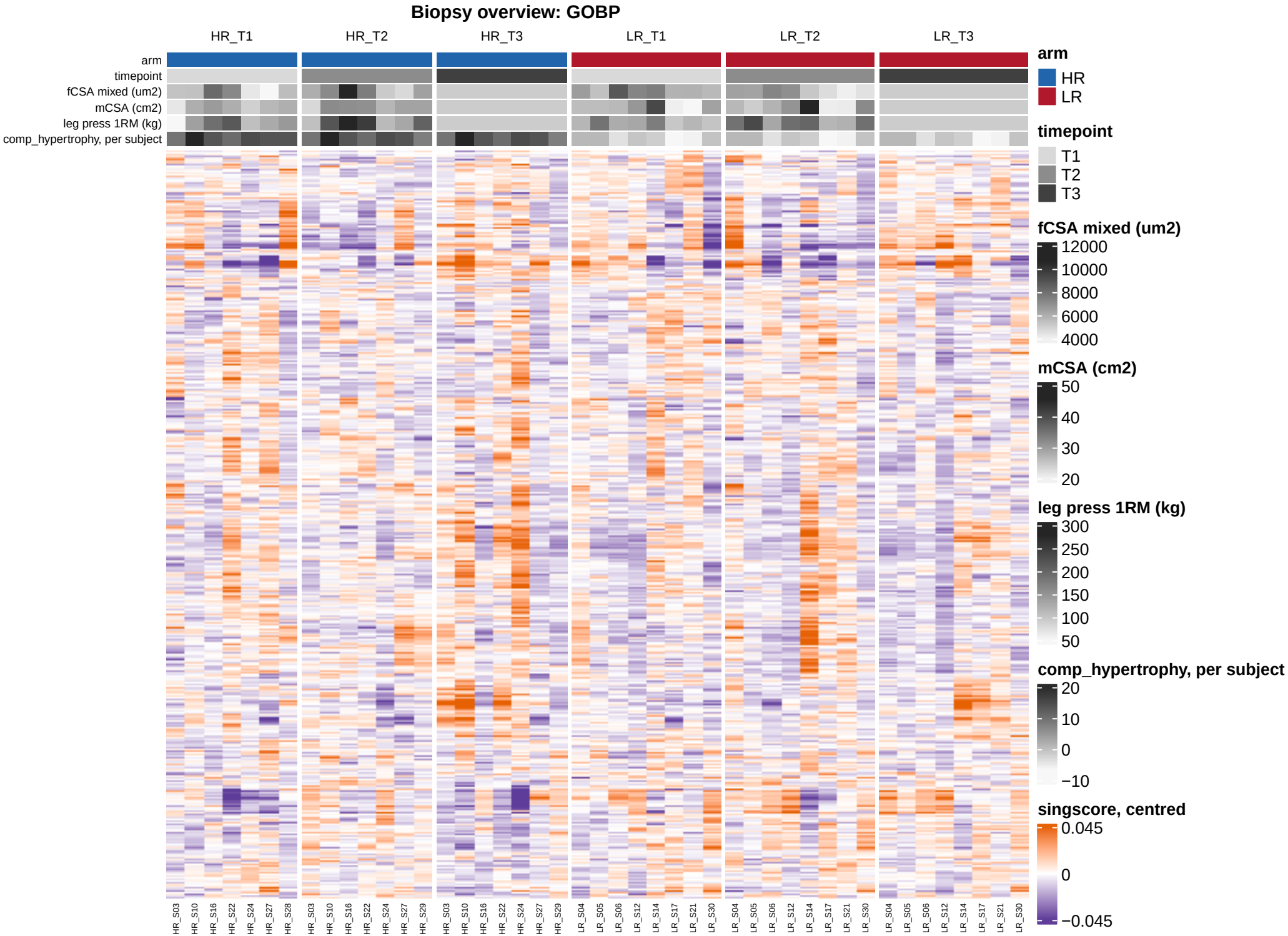


All 62 sets by 45 biopsies, grouped by arm and time point; column labels are subjects. Fill: singscore minus the set's mean over biopsies. Colours stop at the 99th percentile of |value|, 0.053. Rows are ordered within collection by average−linkage clustering (1 − Pearson r) of singscore, for display only. Grey: missing.

Contrast overview: KEGG_Legacy

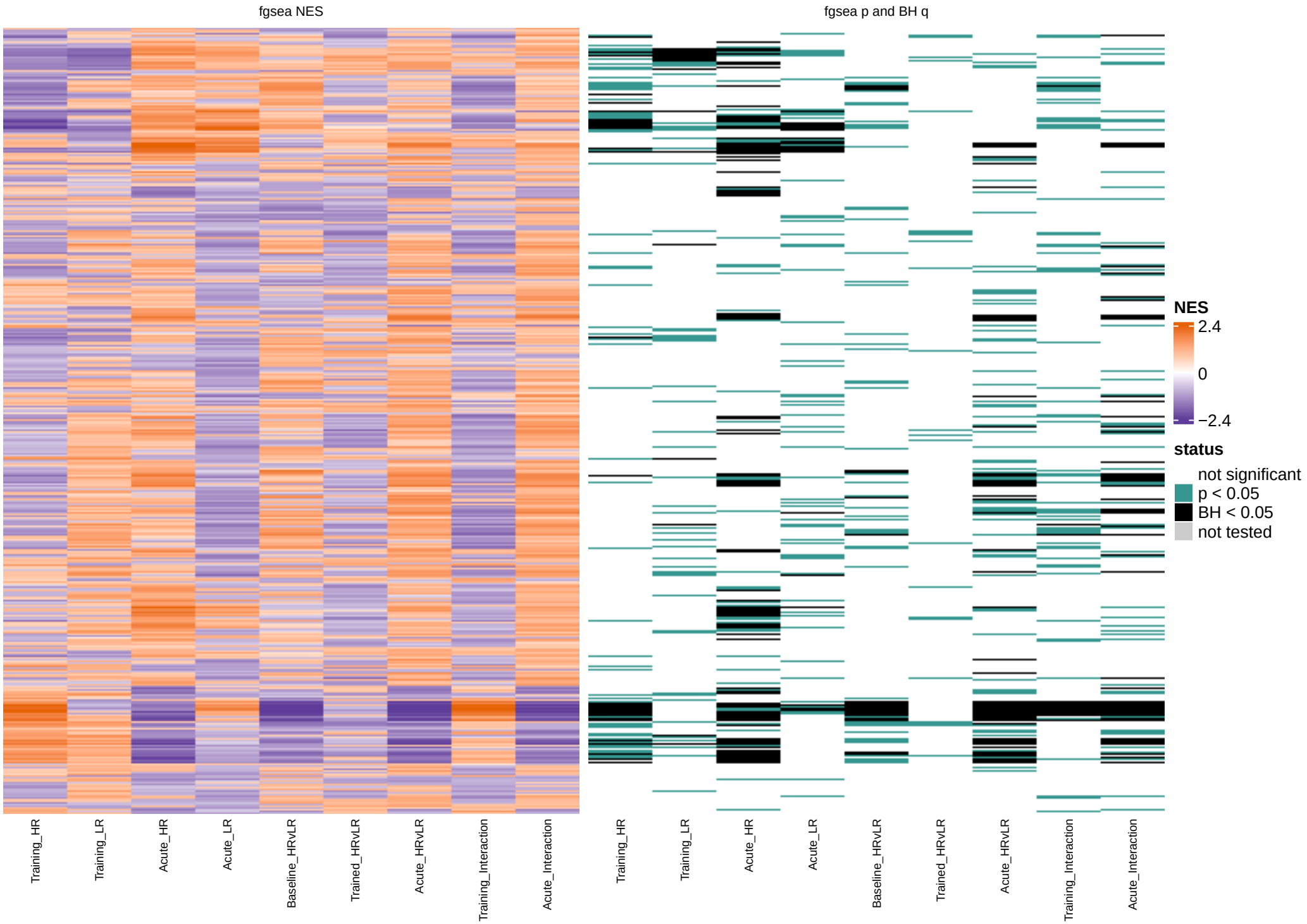


All 62 sets by the nine contrasts. Fill: fgsea normalised enrichment score; status: p and BH q within contrast, collections pooled. Colours stop at the 99th percentile of |value|, 2.4. Rows are ordered within collection by average−linkage clustering (1 − Pearson r) of singscore, for display only. Grey: not tested. Table: c_data/set_tests.rds, set_tests.

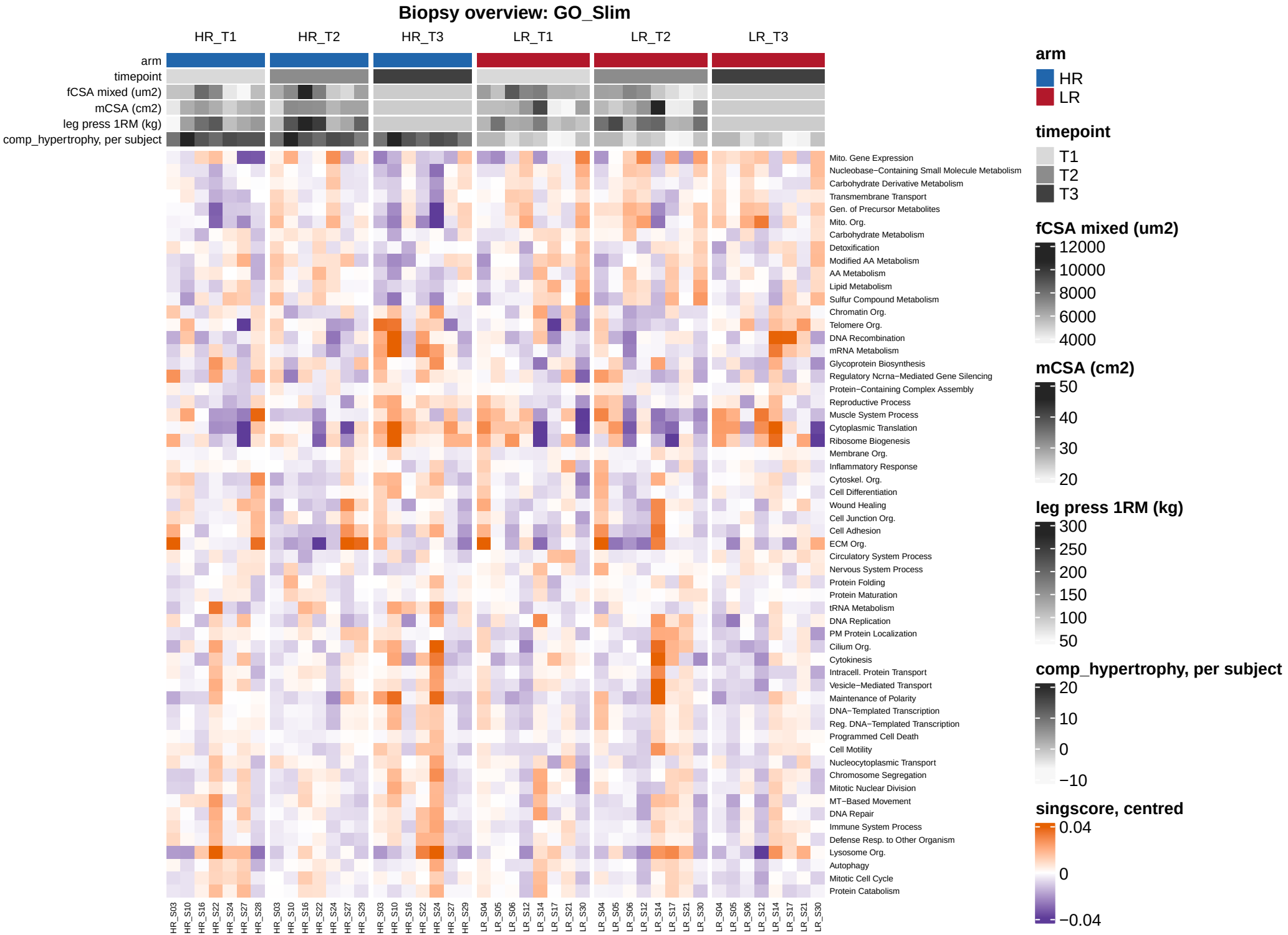


All 897 sets by 45 biopsies, grouped by arm and time point; column labels are subjects. Fill: singscore minus the set's mean over biopsies. Colours stop at the 99th percentile of |value|, 0.045. Rows are ordered within collection by average−linkage clustering (1 − Pearson r) of singscore, for display only. Grey: missing.

Contrast overview: GOBP

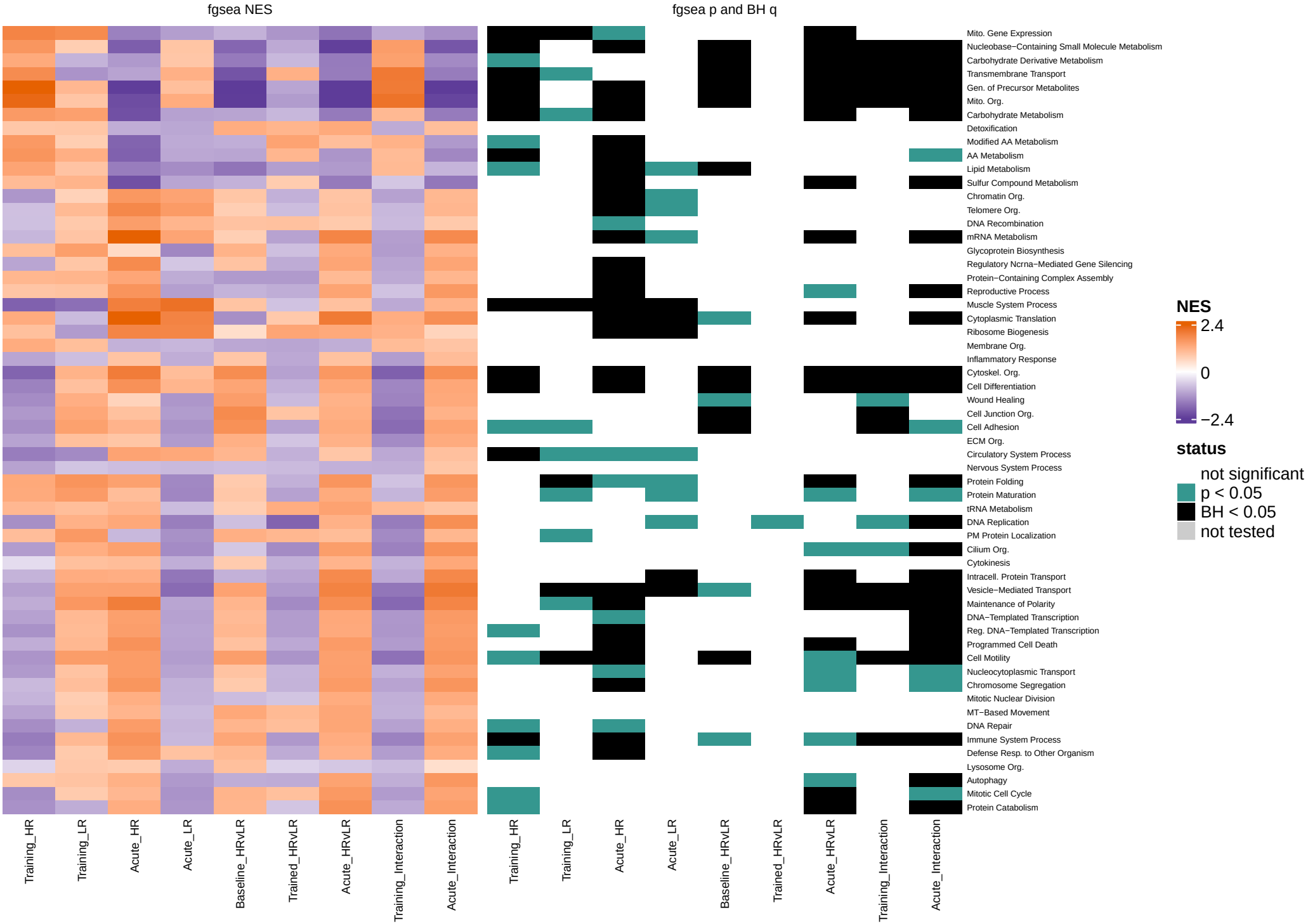


All 897 sets by the nine contrasts. Fill: fgsea normalised enrichment score; status: p and BH q within contrast, collections pooled. Colours stop at the 99th percentile of |value|, 2.4. Rows are ordered within collection by average−linkage clustering (1 − Pearson r) of singscore, for display only. Grey: not tested. Table: c_data/set_tests.rds, set_tests.

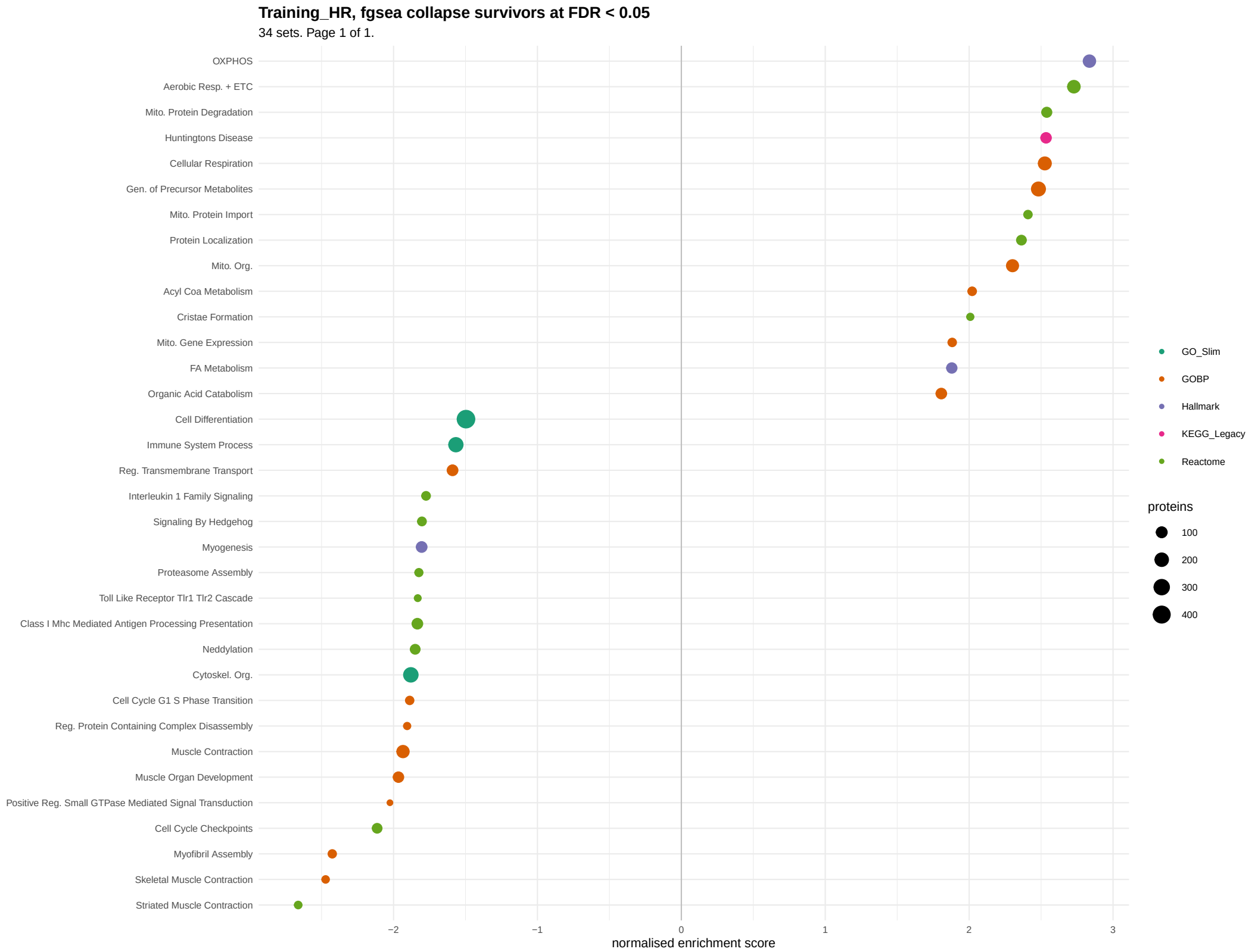


All 58 sets by 45 biopsies, grouped by arm and time point; column labels are subjects. Fill: singscore minus the set's mean over biopsies. Colours stop at the 99th percentile of |value|, 0.04. Rows are ordered within collection by average−linkage clustering (1 − Pearson r) of singscore, for display only. Grey: missing.

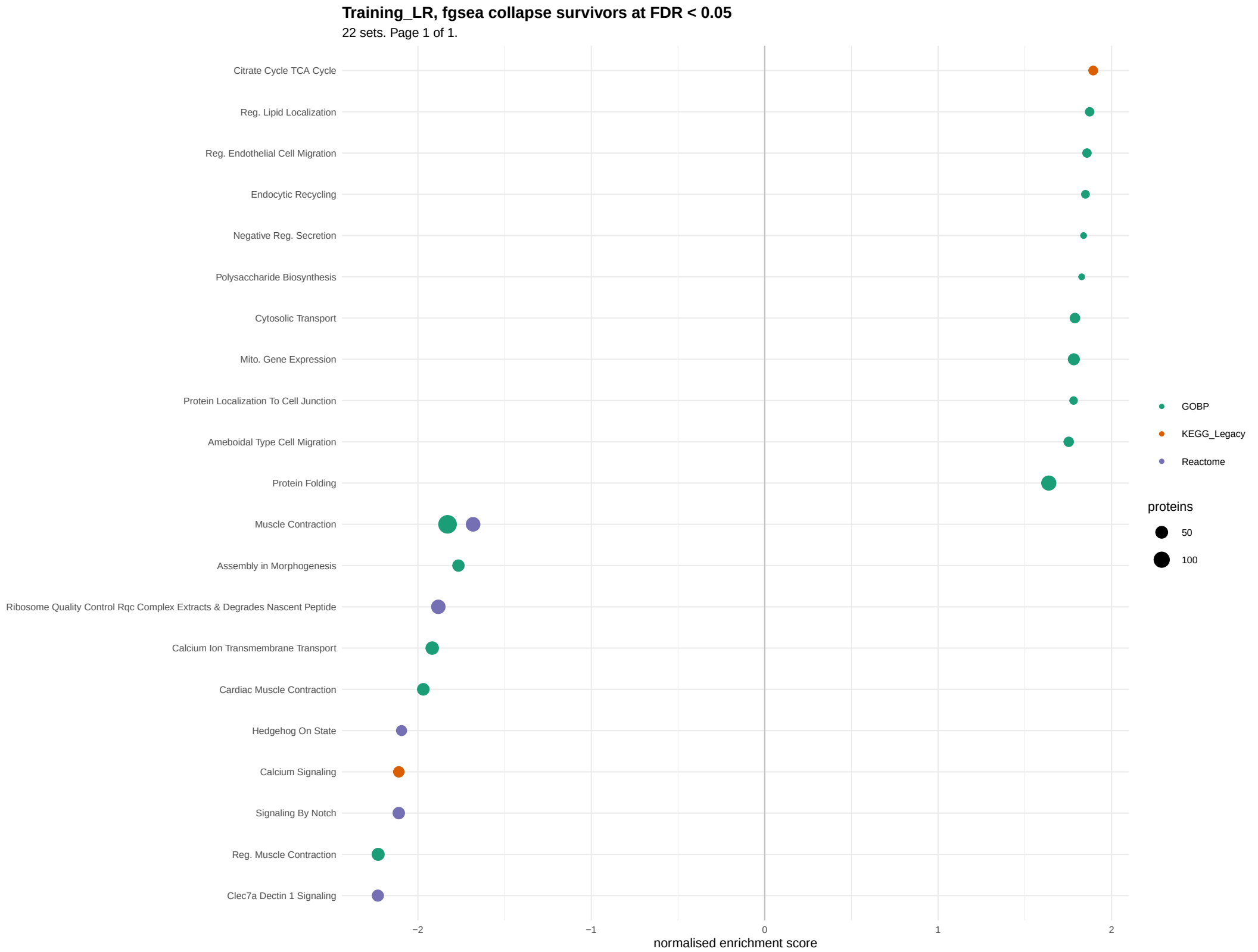
Contrast overview: GO_Slim



All 58 sets by the nine contrasts. Fill: fgsea normalised enrichment score; status: p and BH q within contrast, collections pooled. Colours stop at the 99th percentile of |value|, 2.4. Rows are ordered within collection by average−linkage clustering (1 − Pearson r) of singscore, for display only. Grey: not tested. Table: c_data/set_tests.rds, set_tests.

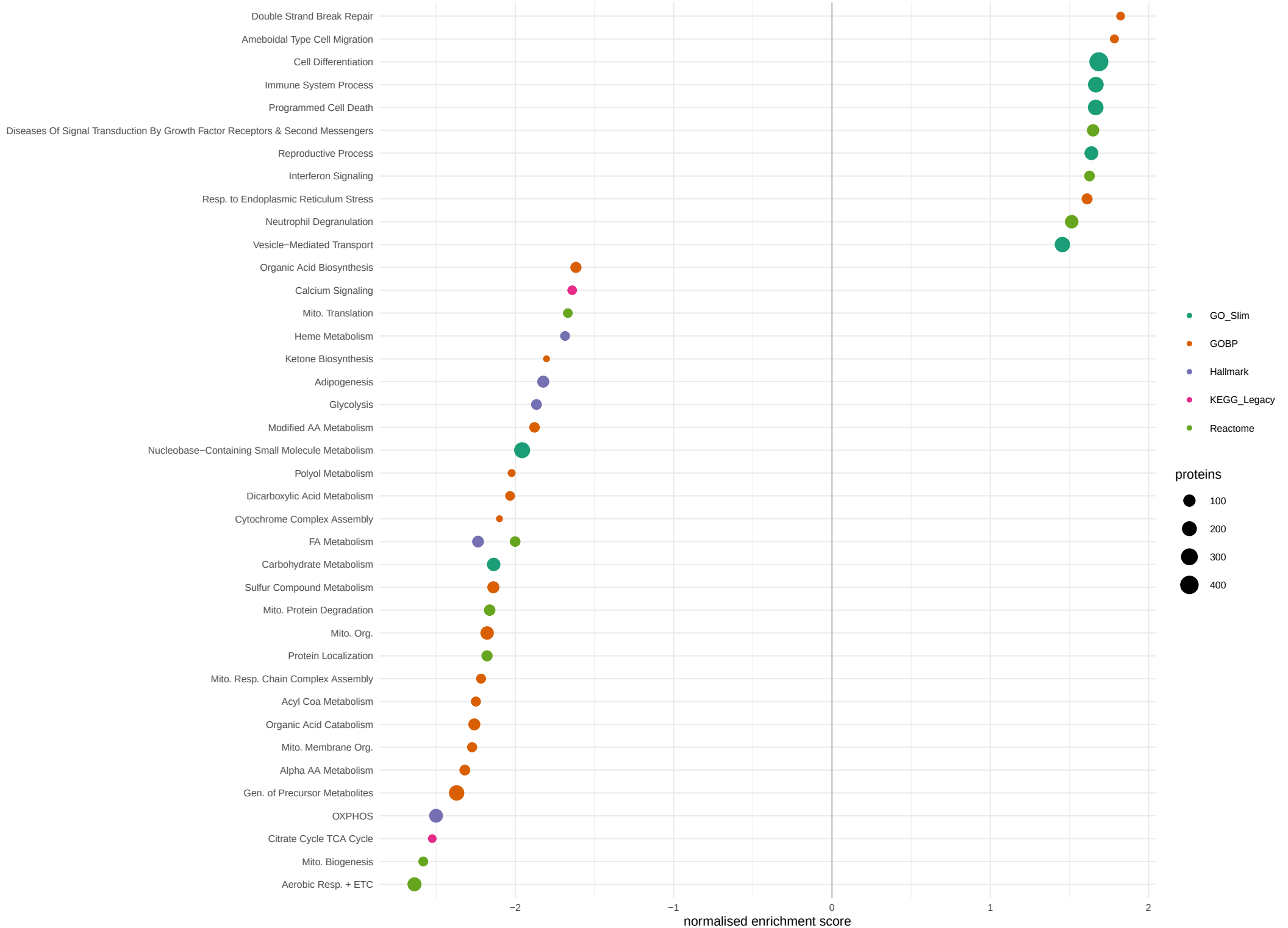


fgsea on the moderated t ranking; BH within contrast over all collections; collapsePathways keeps the sets that stay significant given a stronger set. Table: c_data/02_contrasts.xlsx, significant.

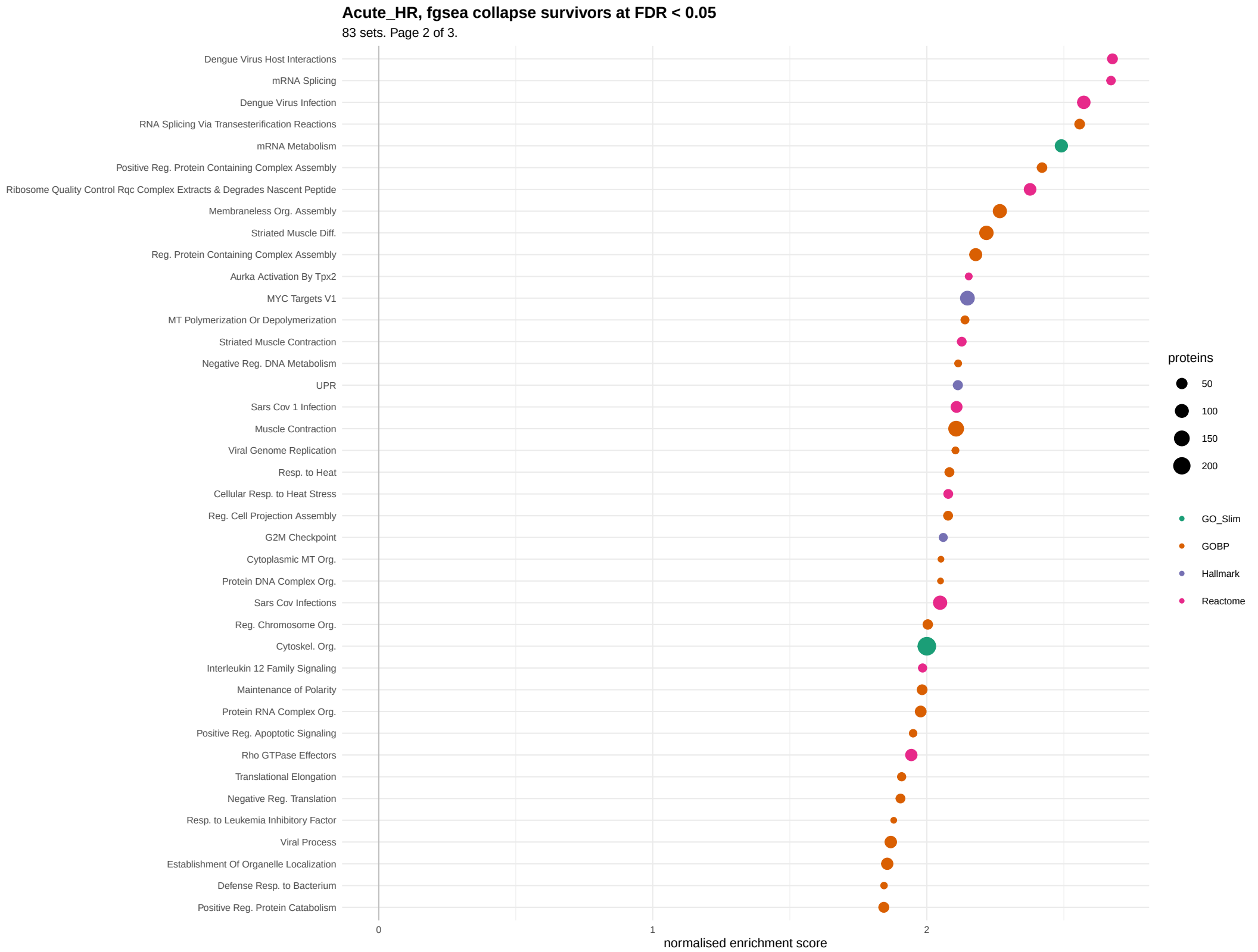

Training_LR, fgsea collapse survivors at FDR < 0.05
22 sets. Page 1 of 1.
Citrate Cycle TCA Cycle
Reg. Lipid Localization
Reg. Endothelial Cell Migration
Endocytic Recycling
Negative Reg. Secretion
Polysaccharide Biosynthesis
Cytosolic Transport
Mito. Gene Expression
Protein Localization To Cell Junction
Ameboidal Type Cell Migration
Protein Folding
Muscle Contraction
Assembly in Morphogenesis
Ribosome Quality Control Rqc Complex Extracts & Degrades Nascent Peptide
Calcium Ion Transmembrane Transport
Cardiac Muscle Contraction
Hedgehog On State
Calcium Signaling
Signaling By Notch
Reg. Muscle Contraction
Clec7a Dectin 1 Signaling
−2
−1
0
1
2
normalised enrichment score
GOBP
KEGG_Legacy
Reactome
proteins
50
100
fgsea on the moderated t ranking; BH within contrast over all collections; collapsePathways keeps the sets that stay significant given a stronger set. Table: c_data/02_contrasts.xlsx, significant.

Acute_HR, fgsea collapse survivors at FDR < 0.05

83 sets. Page 1 of 3.



fgsea on the moderated t ranking; BH within contrast over all collections; collapsePathways keeps the sets that stay significant given a stronger set. Table: c_data/02_contrasts.xlsx, significant.



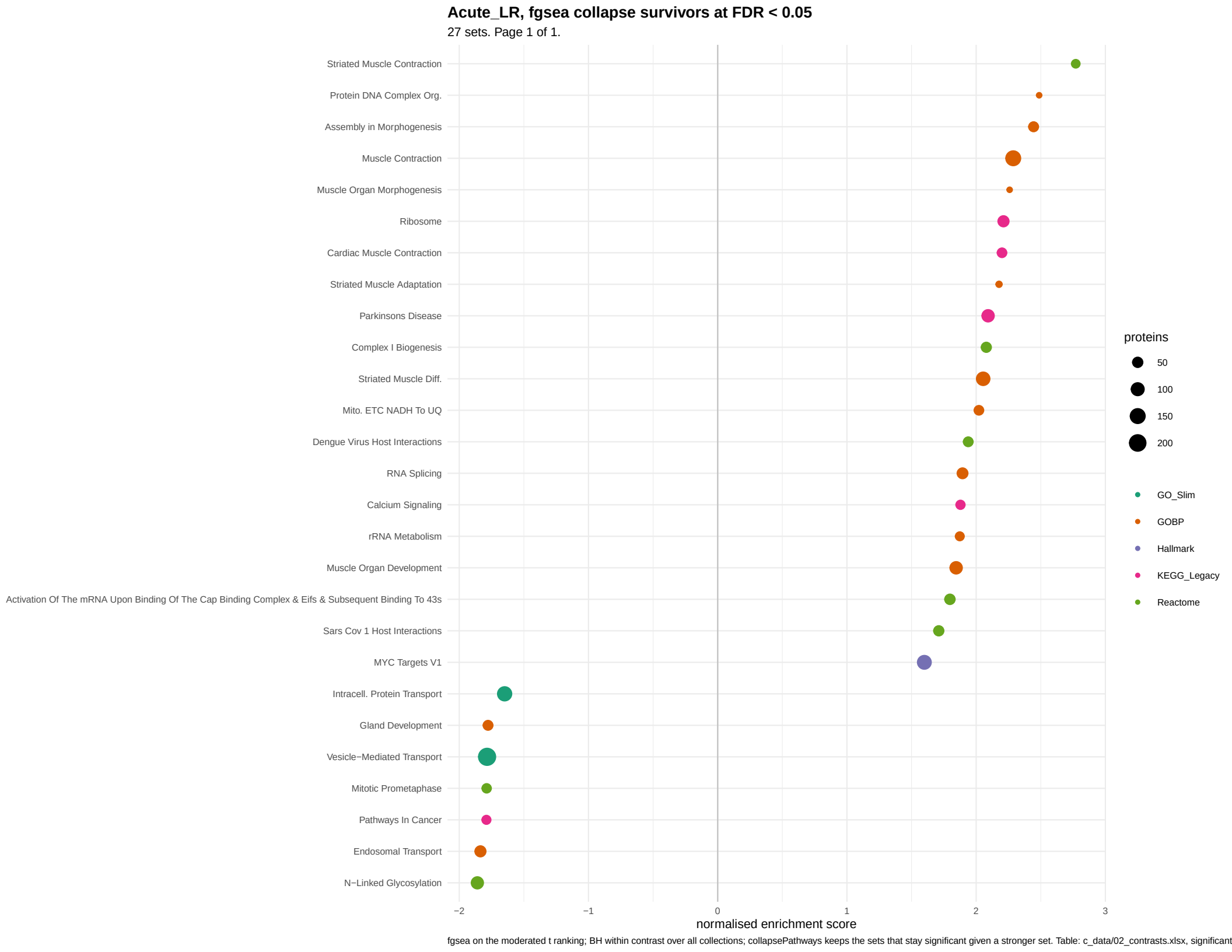
fgsea on the moderated t ranking; BH within contrast over all collections; collapsePathways keeps the sets that stay significant given a stronger set. Table: c_data/02_contrasts.xlsx, significant.

Acute_HR, fgsea collapse survivors at FDR < 0.05

83 sets. Page 3 of 3.

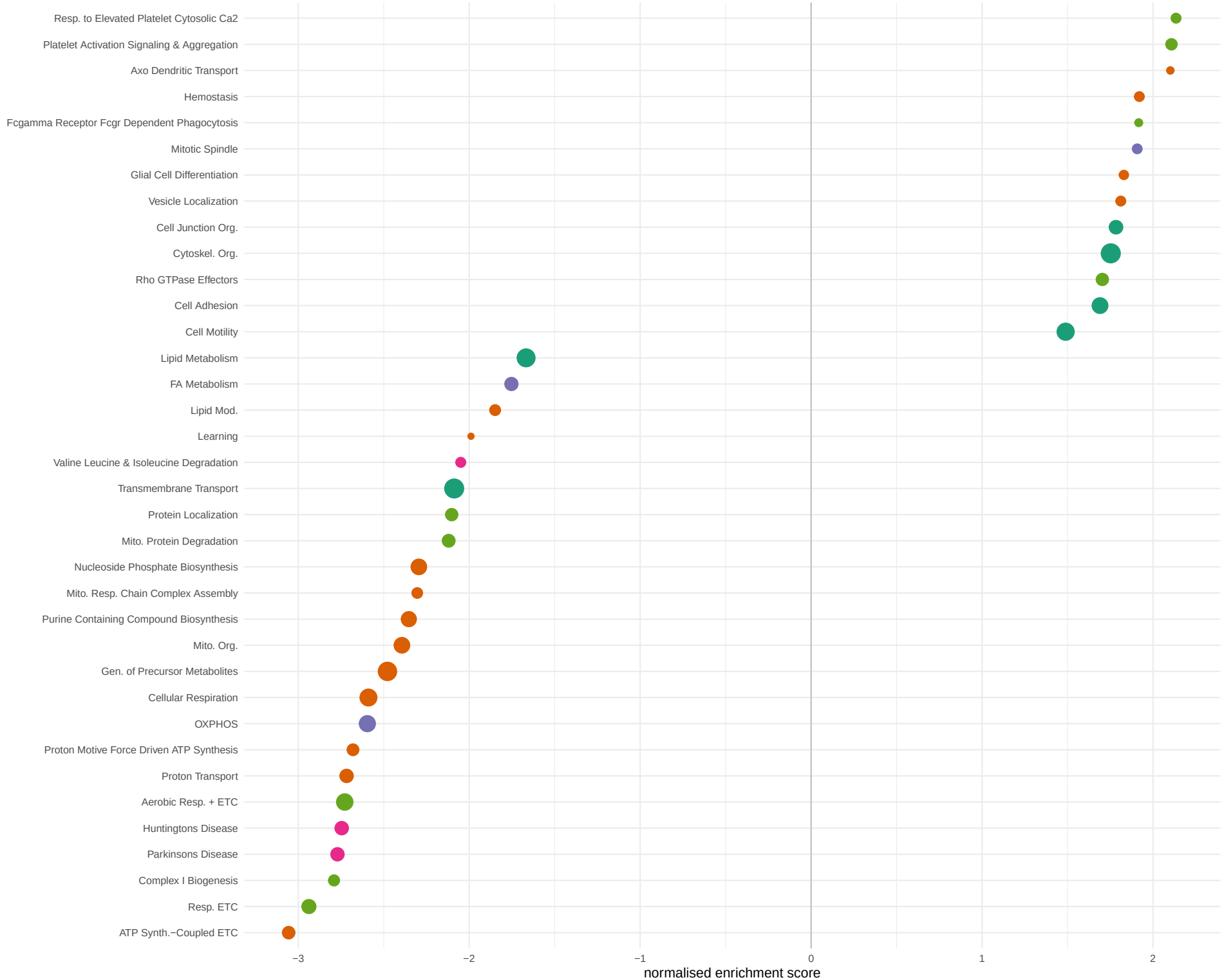


fgsea on the moderated t ranking; BH within contrast over all collections; collapsePathways keeps the sets that stay significant given a stronger set. Table: c_data/02_contrasts.xlsx, significant.


Acute_LR, fgsea collapse survivors at FDR < 0.05
27 sets. Page 1 of 1.
Striated Muscle Contraction
Protein DNA Complex Org.
Assembly in Morphogenesis
Muscle Contraction
Muscle Organ Morphogenesis
Ribosome
Cardiac Muscle Contraction
Striated Muscle Adaptation
Parkinsons Disease
Complex I Biogenesis
Striated Muscle Diff.
Mito. ETC NADH To UQ
Dengue Virus Host Interactions
RNA Splicing
Calcium Signaling
rRNA Metabolism
Muscle Organ Development
Activation Of The mRNA Upon Binding Of The Cap Binding Complex & Eifs & Subsequent Binding To 43s
Sars Cov 1 Host Interactions
MYC Targets V1
Intracell. Protein Transport
Gland Development
Vesicle−Mediated Transport
Mitotic Prometaphase
Pathways In Cancer
Endosomal Transport
N−Linked Glycosylation
−2
−1
0
1
2
3
normalised enrichment score
proteins
50
100
150
200
GO_Slim
GOBP
Hallmark
KEGG_Legacy
Reactome
fgsea on the moderated t ranking; BH within contrast over all collections; collapsePathways keeps the sets that stay significant given a stronger set. Table: c_data/02_contrasts.xlsx, significant

Baseline_HRvLR, fgsea collapse survivors at FDR < 0.05

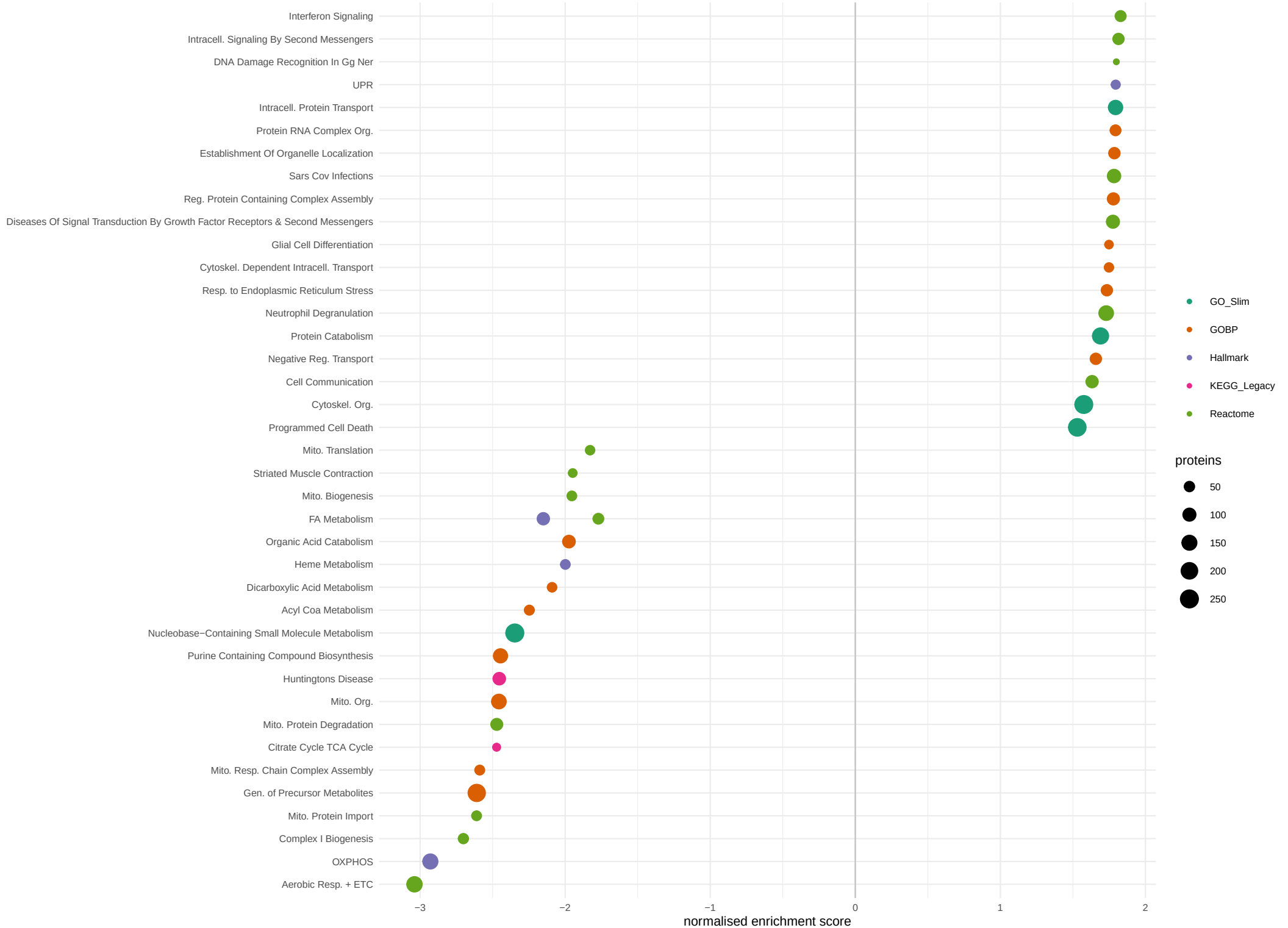
36 sets. Page 1 of 1.



fgsea on the moderated t ranking; BH within contrast over all collections; collapsePathways keeps the sets that stay significant given a stronger set. Table: c_data/02_contrasts.xlsx, significant.

Acute_HRvLR, fgsea collapse survivors at FDR < 0.05

72 sets. Page 1 of 2.



fgsea on the moderated t ranking; BH within contrast over all collections; collapsePathways keeps the sets that stay significant given a stronger set. Table: c_data/02_contrasts.xlsx, significant.

Acute_HRvLR, fgsea collapse survivors at FDR < 0.05

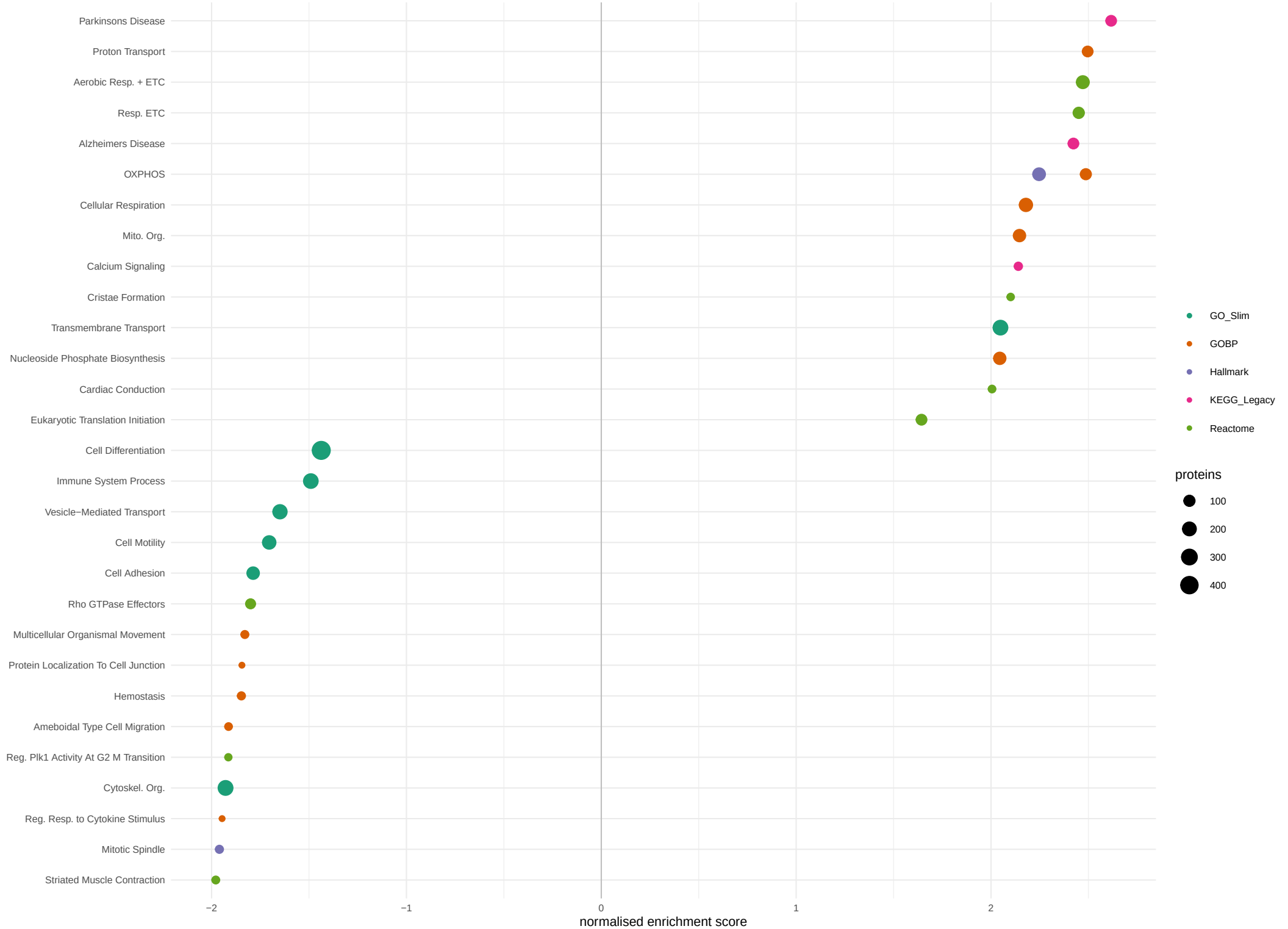
72 sets. Page 2 of 2.



fgsea on the moderated t ranking; BH within contrast over all collections; collapsePathways keeps the sets that stay significant given a stronger set. Table: c_data/02_contrasts.xlsx, significant.

Training_Interaction, fgsea collapse survivors at FDR < 0.05

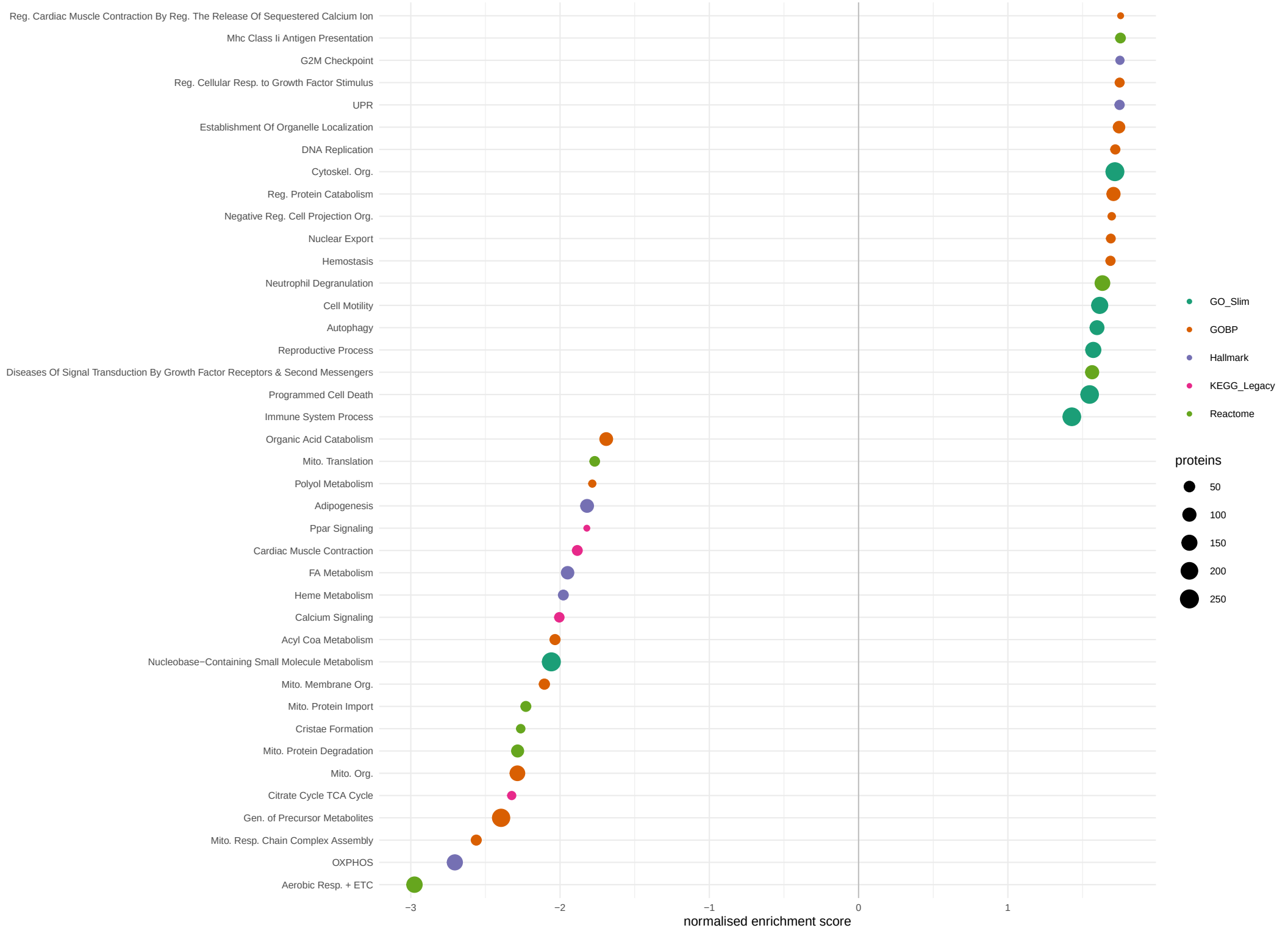
30 sets. Page 1 of 1.



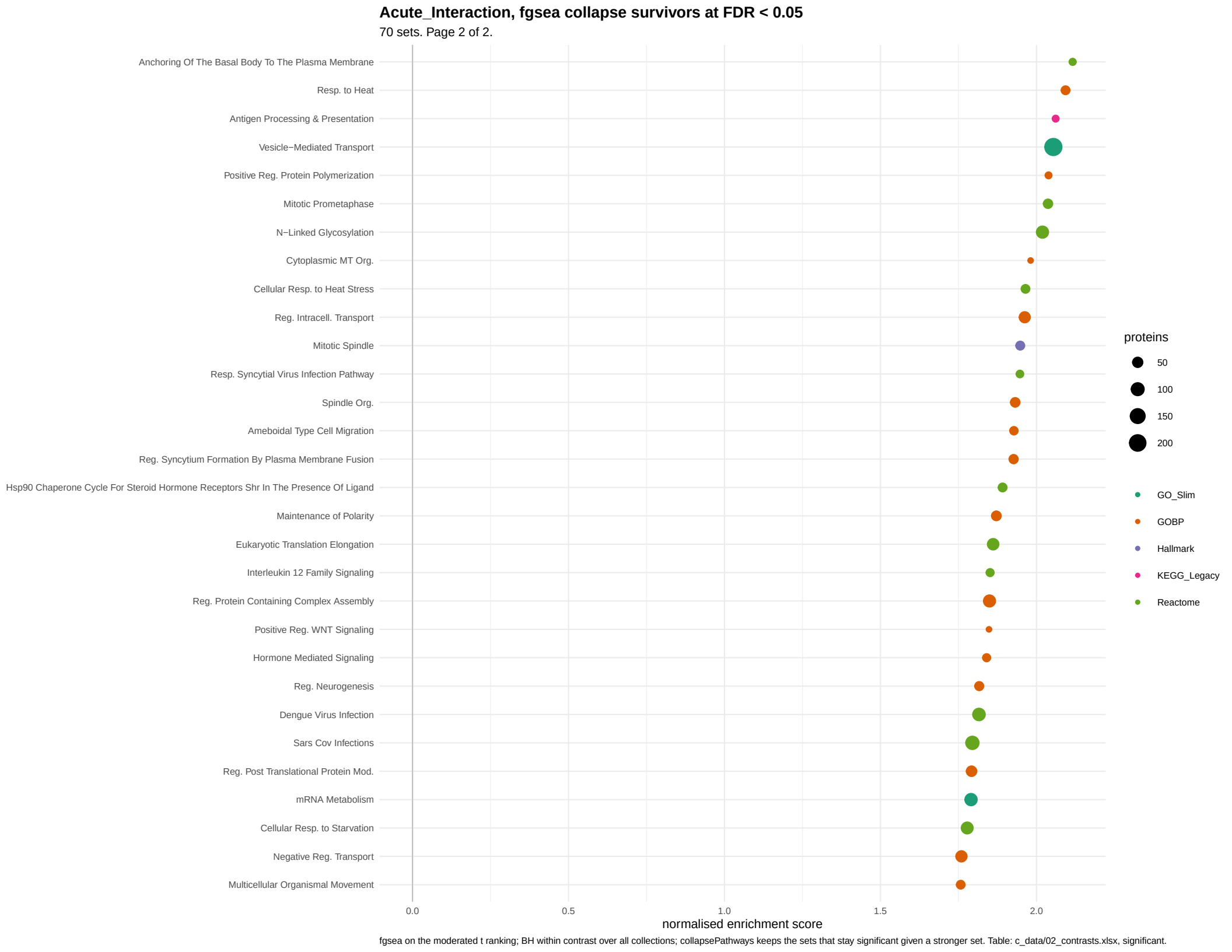
fgsea on the moderated t ranking; BH within contrast over all collections; collapsePathways keeps the sets that stay significant given a stronger set. Table: c_data/02_contrasts.xlsx, significant.

Acute_Interaction, fgsea collapse survivors at FDR < 0.05

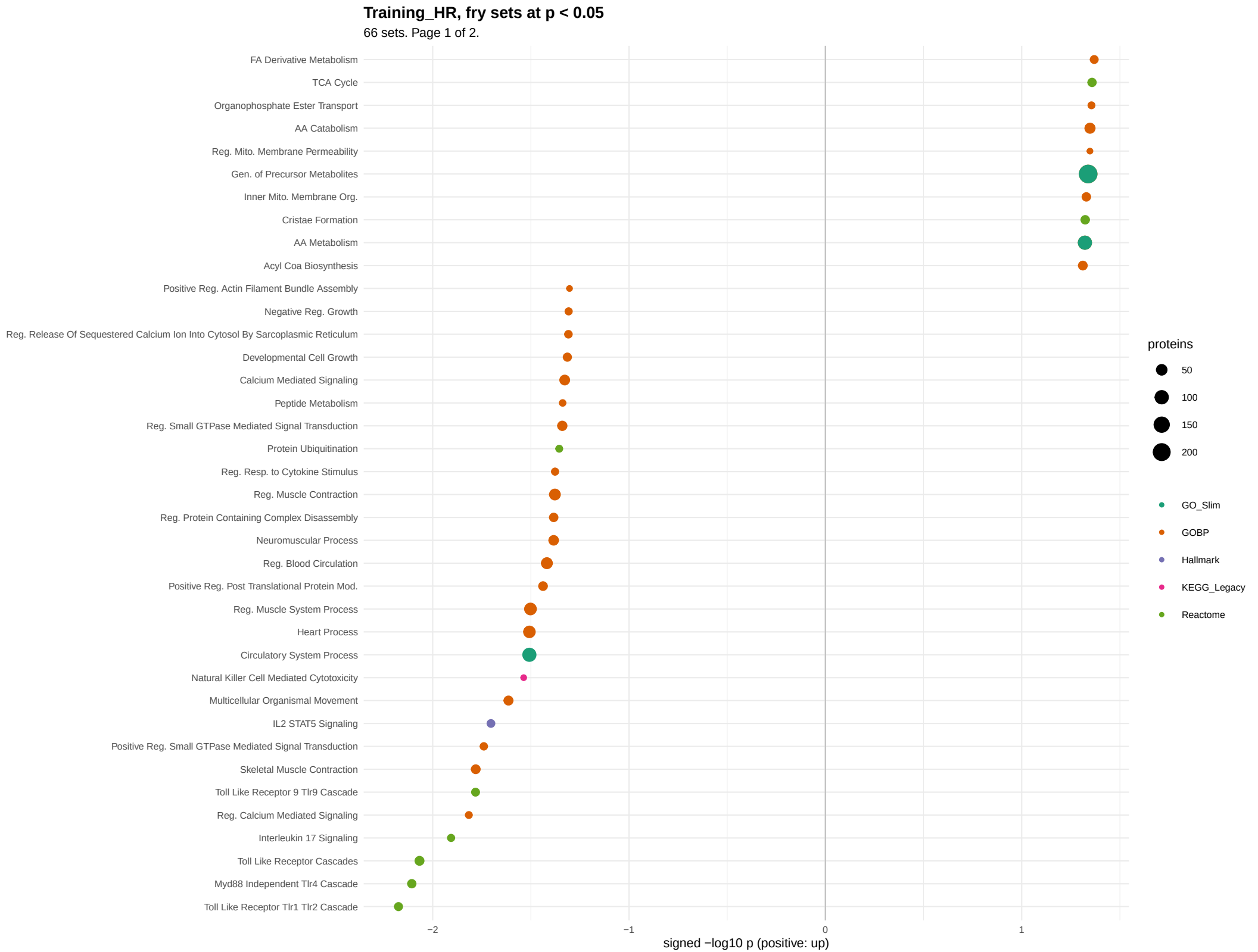
70 sets. Page 1 of 2.



fgsea on the moderated t ranking; BH within contrast over all collections; collapsePathways keeps the sets that stay significant given a stronger set. Table: c_data/02_contrasts.xlsx, significant.

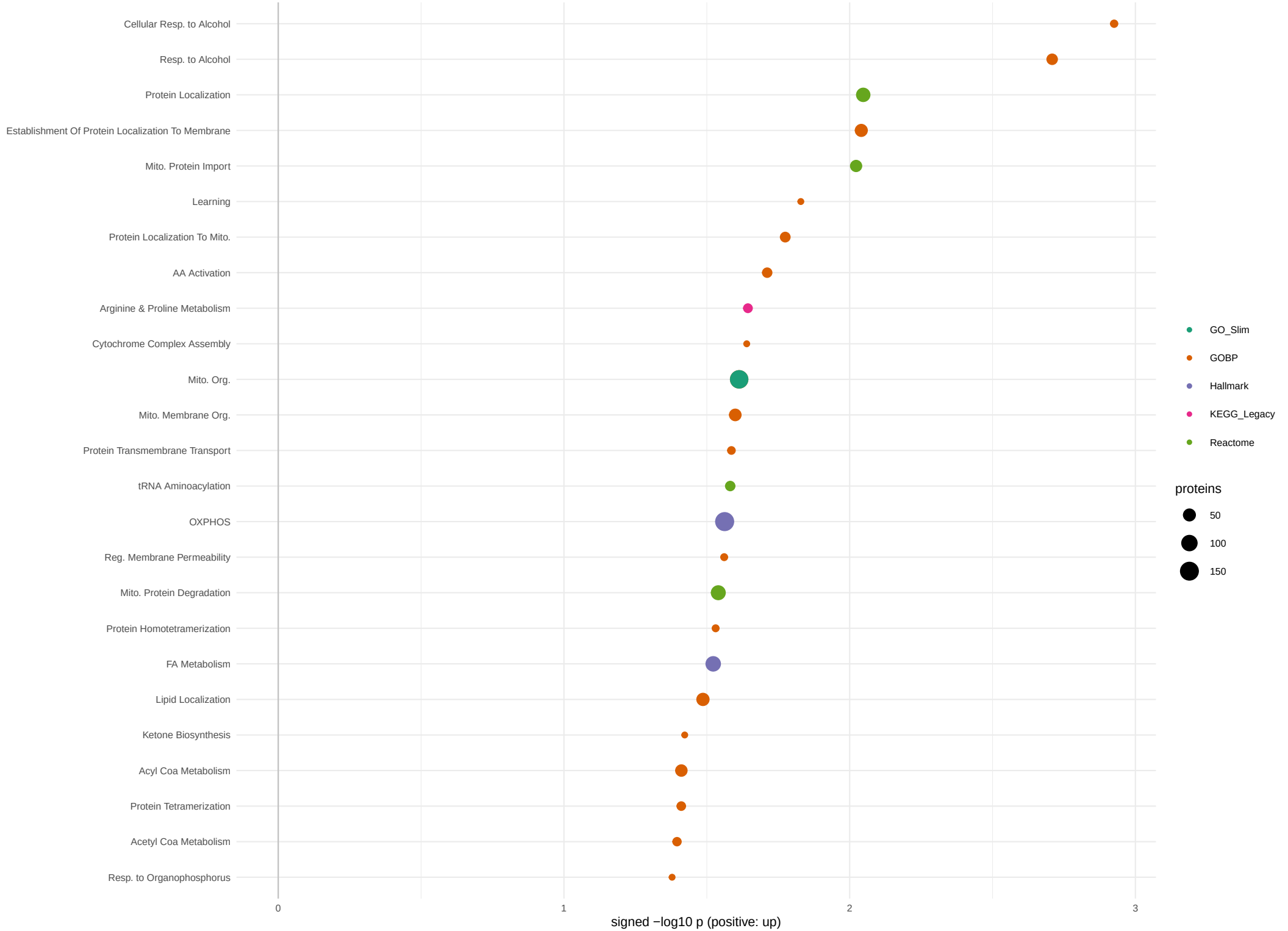


fgsea on the moderated t ranking; BH within contrast over all collections; collapsePathways keeps the sets that stay significant given a stronger set. Table: c_data/02_contrasts.xlsx, significant.

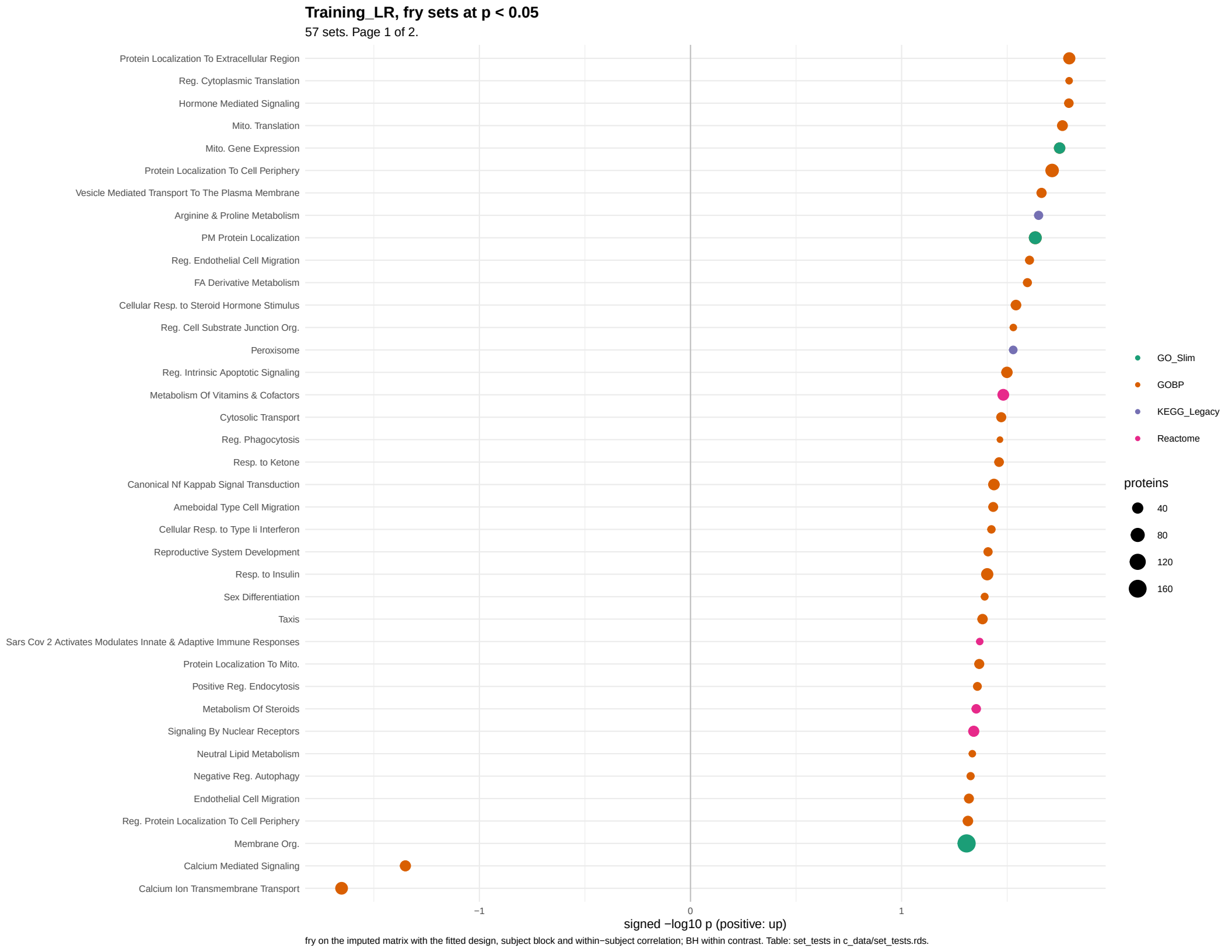

Training_HR, fry sets at p < 0.05
66 sets. Page 1 of 2.
FA Derivative Metabolism
TCA Cycle
Organophosphate Ester Transport
AA Catabolism
Reg. Mito. Membrane Permeability
Gen. of Precursor Metabolites
Inner Mito. Membrane Org.
Cristae Formation
AA Metabolism
Acyl Coa Biosynthesis
Positive Reg. Actin Filament Bundle Assembly
Negative Reg. Growth
Reg. Release Of Sequestered Calcium Ion Into Cytosol By Sarcoplasmic Reticulum
Developmental Cell Growth
Calcium Mediated Signaling
Peptide Metabolism
Reg. Small GTPase Mediated Signal Transduction
Protein Ubiquitination
Reg. Resp. to Cytokine Stimulus
Reg. Muscle Contraction
Reg. Protein Containing Complex Disassembly
Neuromuscular Process
Reg. Blood Circulation
Positive Reg. Post Translational Protein Mod.
Reg. Muscle System Process
Heart Process
Circulatory System Process
Natural Killer Cell Mediated Cytotoxicity
Multicellular Organismal Movement
IL2 STAT5 Signaling
Positive Reg. Small GTPase Mediated Signal Transduction
Skeletal Muscle Contraction
Toll Like Receptor 9 Tlr9 Cascade
Reg. Calcium Mediated Signaling
Interleukin 17 Signaling
Toll Like Receptor Cascades
Myd88 Independent Tlr4 Cascade
Toll Like Receptor Tlr1 Tlr2 Cascade
−2
−1
0
1
signed −log10 p (positive: up)
proteins
50
100
150
200
GO_Slim
GOBP
Hallmark
KEGG_Legacy
Reactome
fry on the imputed matrix with the fitted design, subject block and within−subject correlation; BH within contrast. Table: set_tests in c_data/set_tests.rds.

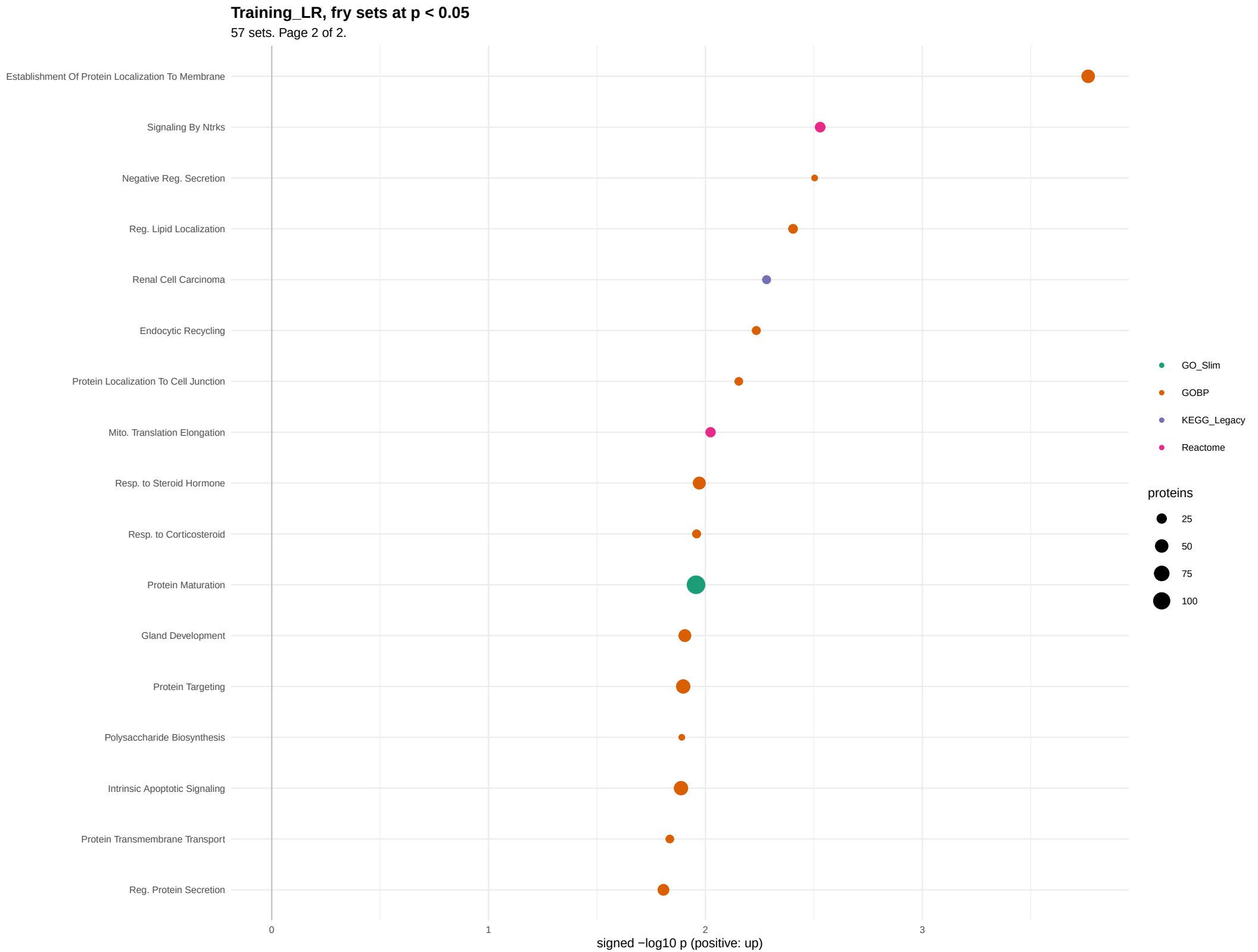
Training_HR, fry sets at p < 0.05

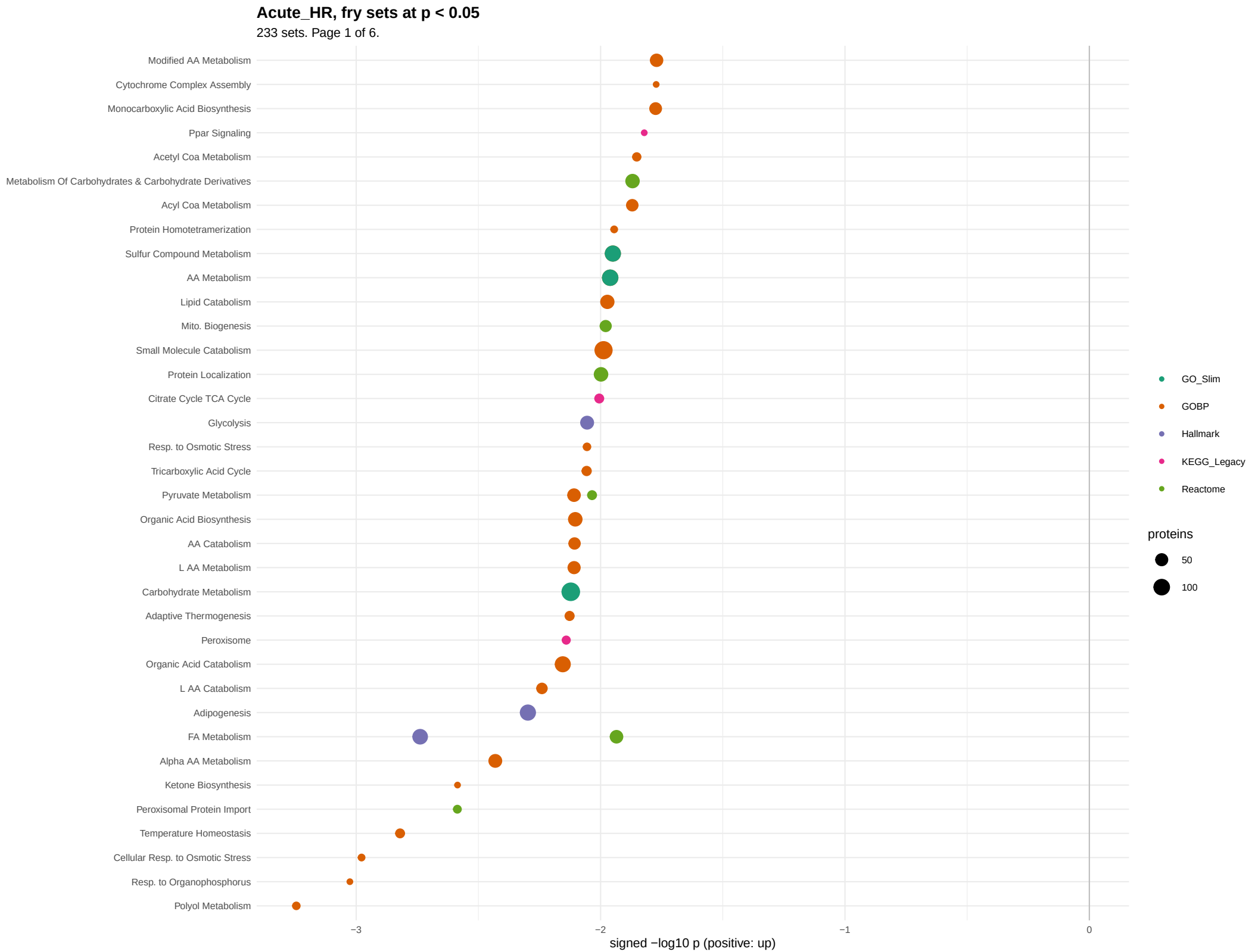
66 sets. Page 2 of 2.



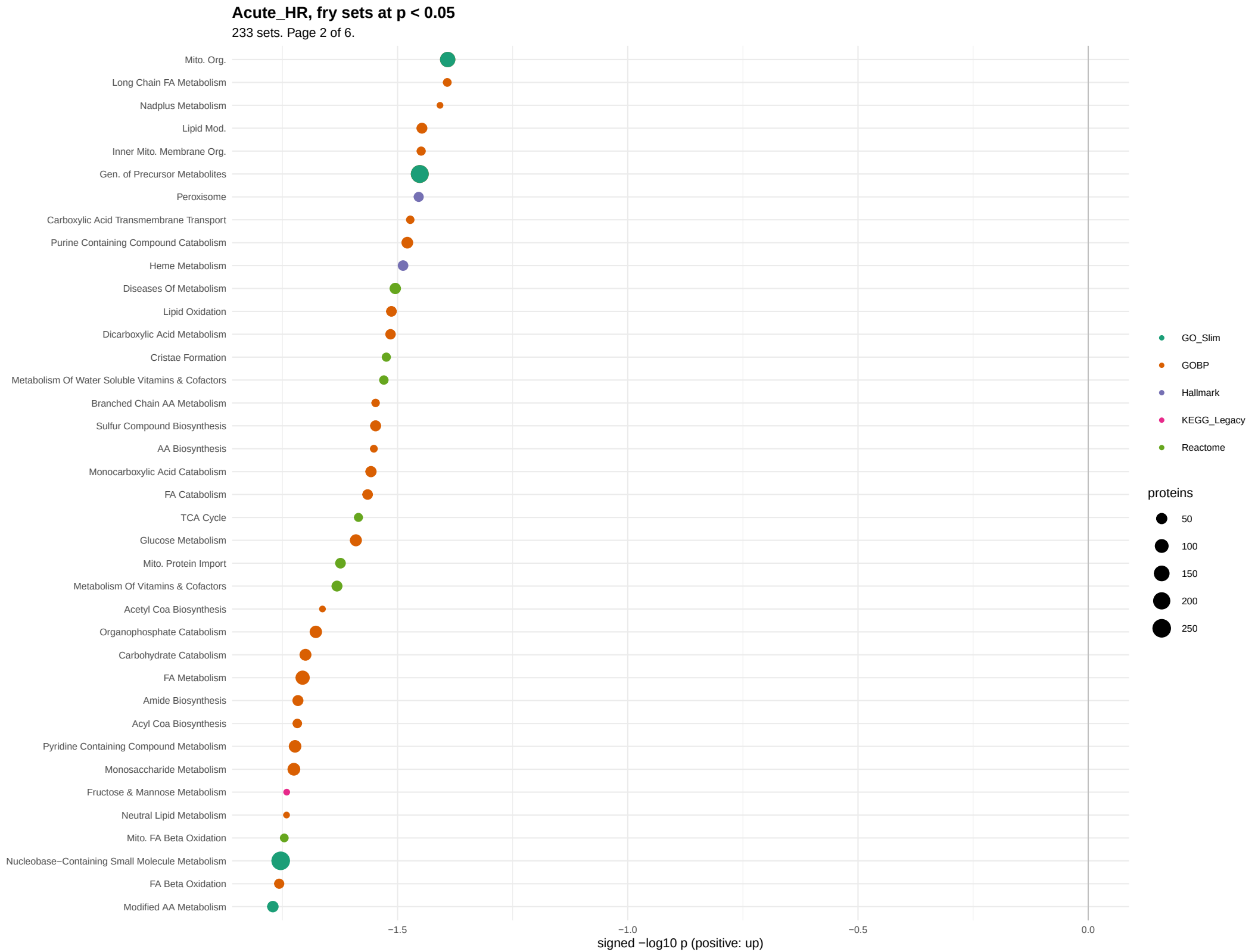
fry on the imputed matrix with the fitted design, subject block and within−subject correlation; BH within contrast. Table: set_tests in c_data/set_tests.rds.


Training_LR, fry sets at p < 0.05
57 sets. Page 1 of 2.
Protein Localization To Extracellular Region
Reg. Cytoplasmic Translation
Hormone Mediated Signaling
Mito. Translation
Mito. Gene Expression
Protein Localization To Cell Periphery
Vesicle Mediated Transport To The Plasma Membrane
Arginine & Proline Metabolism
PM Protein Localization
Reg. Endothelial Cell Migration
FA Derivative Metabolism
Cellular Resp. to Steroid Hormone Stimulus
Reg. Cell Substrate Junction Org.
Peroxisome
Reg. Intrinsic Apoptotic Signaling
Metabolism Of Vitamins & Cofactors
Cytosolic Transport
Reg. Phagocytosis
Resp. to Ketone
Canonical Nf Kappab Signal Transduction
Ameboidal Type Cell Migration
Cellular Resp. to Type Ii Interferon
Reproductive System Development
Resp. to Insulin
Sex Differentiation
Taxis
Sars Cov 2 Activates Modulates Innate & Adaptive Immune Responses
Protein Localization To Mito.
Positive Reg. Endocytosis
Metabolism Of Steroids
Signaling By Nuclear Receptors
Neutral Lipid Metabolism
Negative Reg. Autophagy
Endothelial Cell Migration
Reg. Protein Localization To Cell Periphery
Membrane Org.
Calcium Mediated Signaling
Calcium Ion Transmembrane Transport
−1
0
1
signed −log10 p (positive: up)
GO_Slim
GOBP
KEGG_Legacy
Reactome
proteins
40
80
120
160
fry on the imputed matrix with the fitted design, subject block and within−subject correlation; BH within contrast. Table: set_tests in c_data/set_tests.rds.

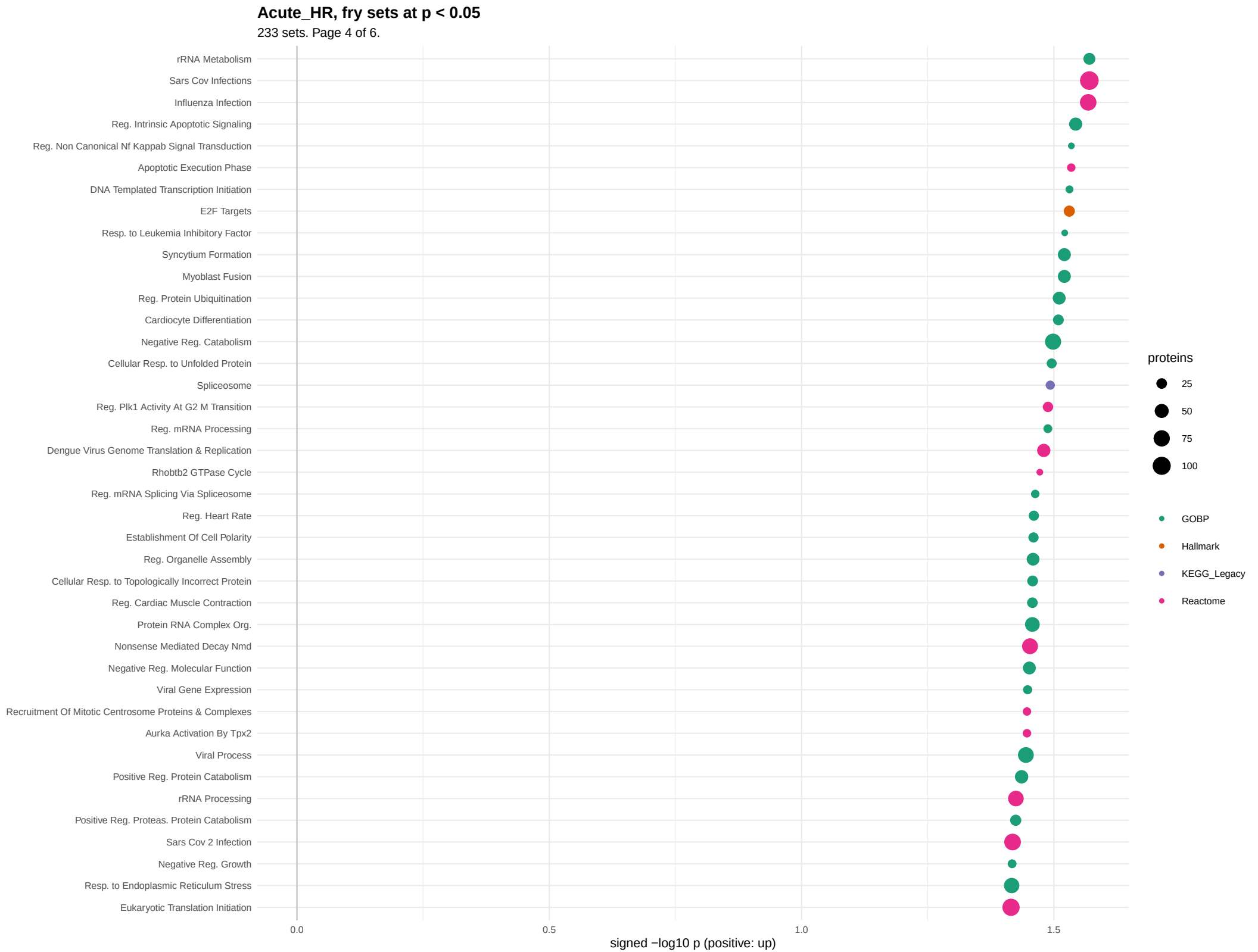

Training_LR, fry sets at p < 0.05
57 sets. Page 2 of 2.
Establishment Of Protein Localization To Membrane
Signaling By Ntrks
Negative Reg. Secretion
Reg. Lipid Localization
Renal Cell Carcinoma
Endocytic Recycling
Protein Localization To Cell Junction
Mito. Translation Elongation
Resp. to Steroid Hormone
Resp. to Corticosteroid
Protein Maturation
Gland Development
Protein Targeting
Polysaccharide Biosynthesis
Intrinsic Apoptotic Signaling
Protein Transmembrane Transport
Reg. Protein Secretion
0
1
2
3
signed −log10 p (positive: up)
GO_Slim
GOBP
KEGG_Legacy
Reactome
proteins
25
50
75
100
fry on the imputed matrix with the fitted design, subject block and within−subject correlation; BH within contrast. Table: set_tests in c_data/set_tests.rds.

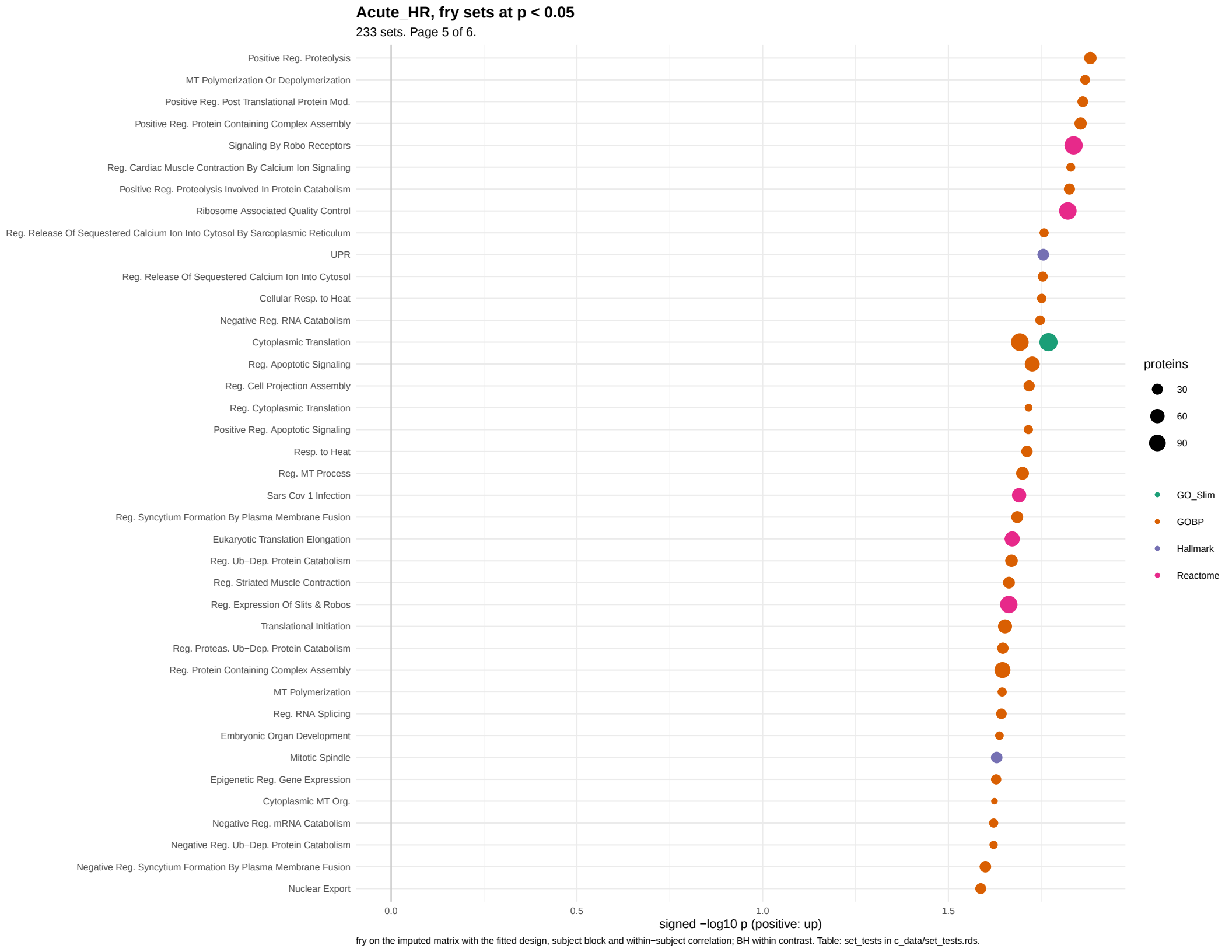


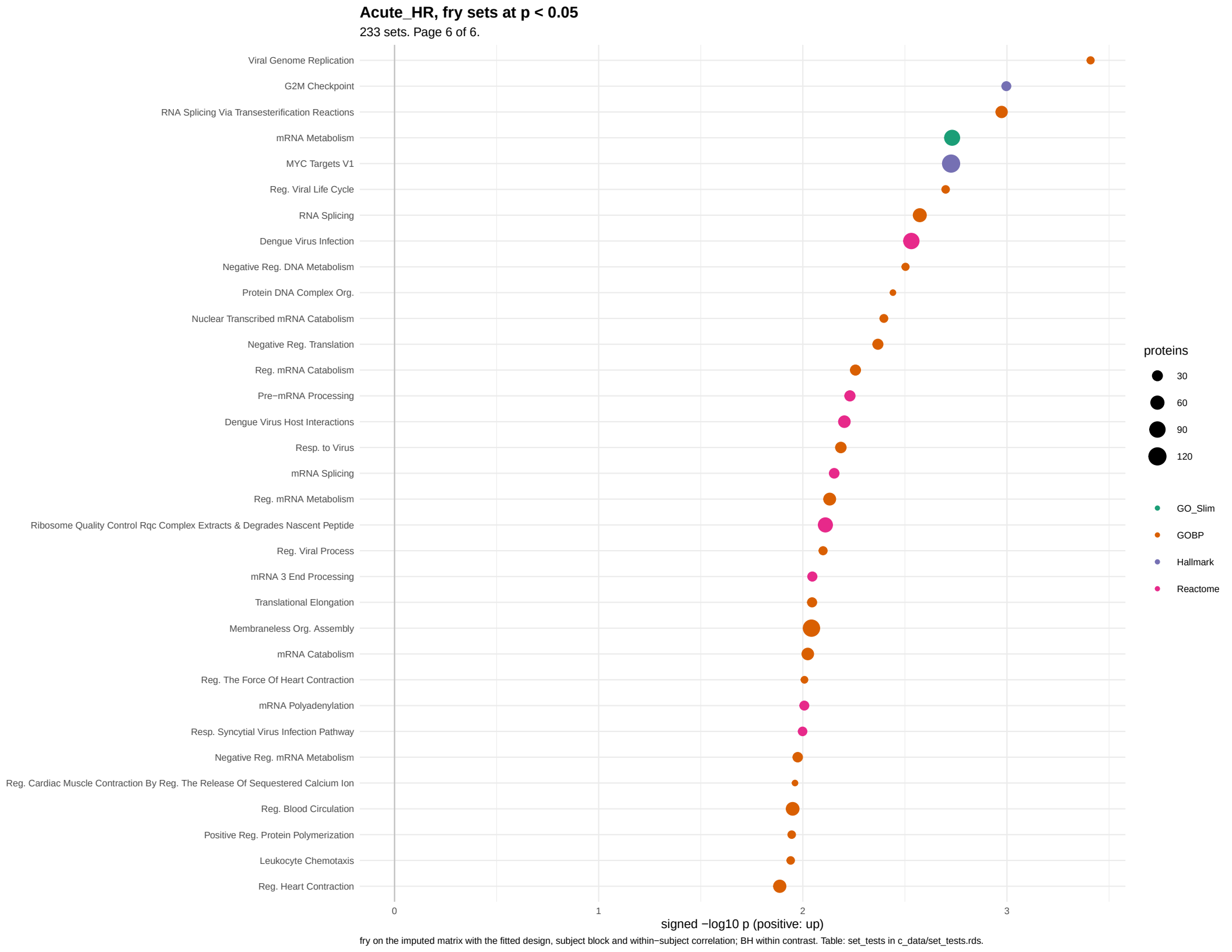
fry on the imputed matrix with the fitted design, subject block and within−subject correlation; BH within contrast. Table: set_tests in c_data/set_tests.rds.

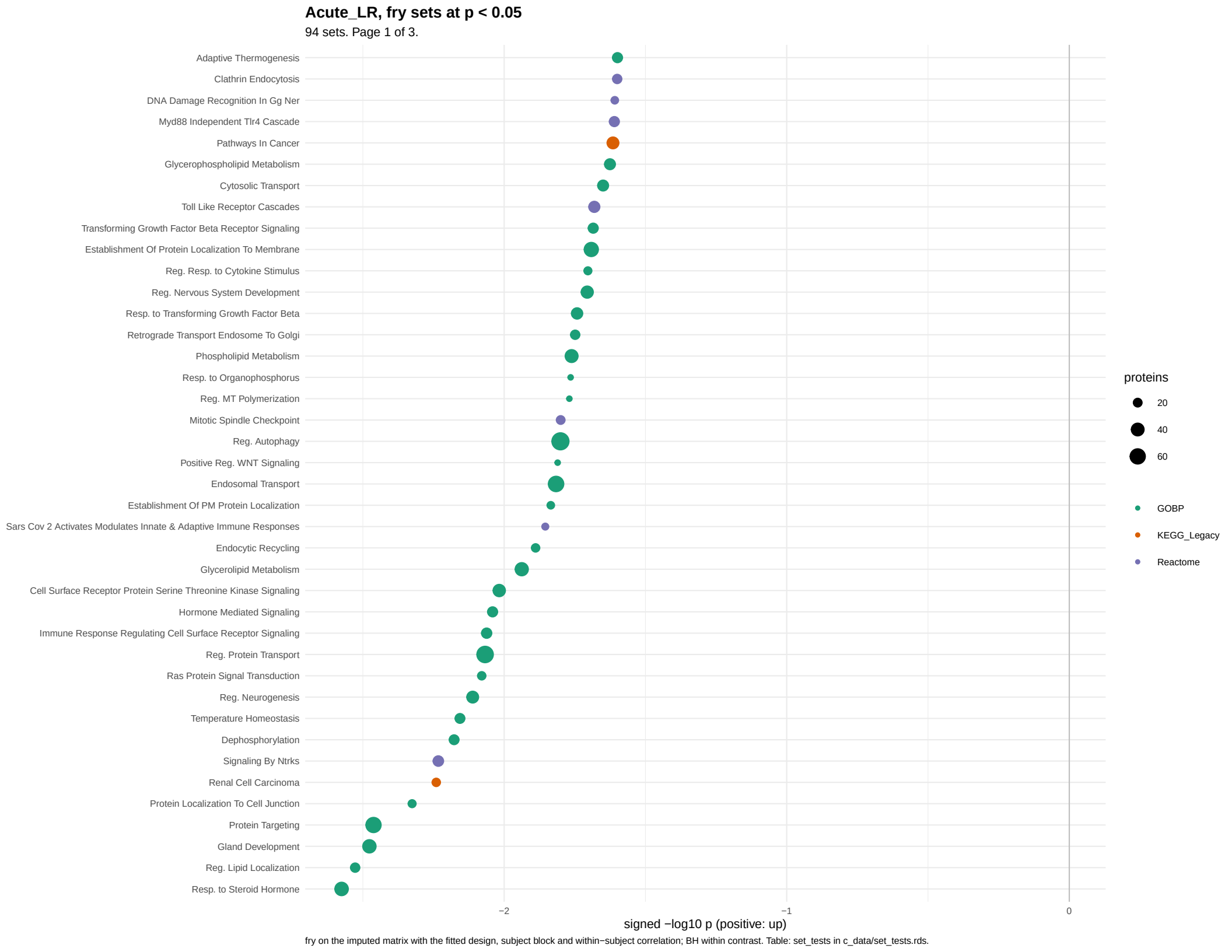

Acute_HR, fry sets at p < 0.05
233 sets. Page 2 of 6.
Mito. Org.
Long Chain FA Metabolism
Nadplus Metabolism
Lipid Mod.
Inner Mito. Membrane Org.
Gen. of Precursor Metabolites
Peroxisome
Carboxylic Acid Transmembrane Transport
Purine Containing Compound Catabolism
Heme Metabolism
Diseases Of Metabolism
Lipid Oxidation
Dicarboxylic Acid Metabolism
Cristae Formation
Metabolism Of Water Soluble Vitamins & Cofactors
Branched Chain AA Metabolism
Sulfur Compound Biosynthesis
AA Biosynthesis
Monocarboxylic Acid Catabolism
FA Catabolism
TCA Cycle
Glucose Metabolism
Mito. Protein Import
Metabolism Of Vitamins & Cofactors
Acetyl Coa Biosynthesis
Organophosphate Catabolism
Carbohydrate Catabolism
FA Metabolism
Amide Biosynthesis
Acyl Coa Biosynthesis
Pyridine Containing Compound Metabolism
Monosaccharide Metabolism
Fructose & Mannose Metabolism
Neutral Lipid Metabolism
Mito. FA Beta Oxidation
Nucleobase−Containing Small Molecule Metabolism
FA Beta Oxidation
Modified AA Metabolism
−1.5
−1.0
−0.5
0.0
signed −log10 p (positive: up)
GO_Slim
GOBP
Hallmark
KEGG_Legacy
Reactome
proteins
50
100
150
200
250
fry on the imputed matrix with the fitted design, subject block and within−subject correlation; BH within contrast. Table: set_tests in c_data/set_tests.rds.


Acute_HR, fry sets at p < 0.05
233 sets. Page 3 of 6.
Sars Cov 1 Host Interactions
Double Strand Break Repair
Allograft Rejection
Reg. Protein Polymerization
Resp. to Topologically Incorrect Protein
Protein Polymerization
Defense Resp. to Bacterium
Spindle Org.
Maintenance of Polarity
Striated Muscle Contraction
Sars Cov 2 Host Interactions
Influenza Viral RNA Transcription & Replication
Muscle Hypertrophy
Erad Pathway
Bacterial Infection Pathways
Rho GTPase Effectors
Non Canonical Nf Kappab Signal Transduction
Reg. DNA Recombination
In Utero Embryonic Development
Selenoamino Acid Metabolism
Rho Gtpases Activate Pkns
Establishment Of Organelle Localization
Mitotic Prophase
Cell Chemotaxis
Lamellipodium Assembly
Mitotic Spindle Checkpoint
Intrinsic Apoptotic Signaling
Defense Resp. to Virus
Reg. Resp. to Endoplasmic Reticulum Stress
Cytoskel. Org.
Branched Chain AA Catabolism
Lipid Metabolism
Polysaccharide Catabolism
Mito. Protein Degradation
Alpha AA Biosynthesis
Mito. Membrane Org.
Glycogen Metabolism
Arginine & Proline Metabolism
Protein Tetramerization
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
signed −log10 p (positive: up)
proteins
50
100
150
200
GO_Slim
GOBP
Hallmark
KEGG_Legacy
Reactome
fry on the imputed matrix with the fitted design, subject block and within−subject correlation; BH within contrast. Table: set_tests in c_data/set_tests.rds.


Acute_HR, fry sets at p < 0.05
233 sets. Page 4 of 6.
rRNA Metabolism
Sars Cov Infections
Influenza Infection
Reg. Intrinsic Apoptotic Signaling
Reg. Non Canonical Nf Kappab Signal Transduction
Apoptotic Execution Phase
DNA Templated Transcription Initiation
E2F Targets
Resp. to Leukemia Inhibitory Factor
Syncytium Formation
Myoblast Fusion
Reg. Protein Ubiquitination
Cardiocyte Differentiation
Negative Reg. Catabolism
Cellular Resp. to Unfolded Protein
Spliceosome
Reg. Plk1 Activity At G2 M Transition
Reg. mRNA Processing
Dengue Virus Genome Translation & Replication
Rhobtb2 GTPase Cycle
Reg. mRNA Splicing Via Spliceosome
Reg. Heart Rate
Establishment Of Cell Polarity
Reg. Organelle Assembly
Cellular Resp. to Topologically Incorrect Protein
Reg. Cardiac Muscle Contraction
Protein RNA Complex Org.
Nonsense Mediated Decay Nmd
Negative Reg. Molecular Function
Viral Gene Expression
Recruitment Of Mitotic Centrosome Proteins & Complexes
Aurka Activation By Tpx2
Viral Process
Positive Reg. Protein Catabolism
rRNA Processing
Positive Reg. Proteas. Protein Catabolism
Sars Cov 2 Infection
Negative Reg. Growth
Resp. to Endoplasmic Reticulum Stress
Eukaryotic Translation Initiation
0.0
0.5
1.0
1.5
signed −log10 p (positive: up)
proteins
25
50
75
100
GOBP
Hallmark
KEGG_Legacy
Reactome
fry on the imputed matrix with the fitted design, subject block and within−subject correlation; BH within contrast. Table: set_tests in c_data/set_tests.rds.

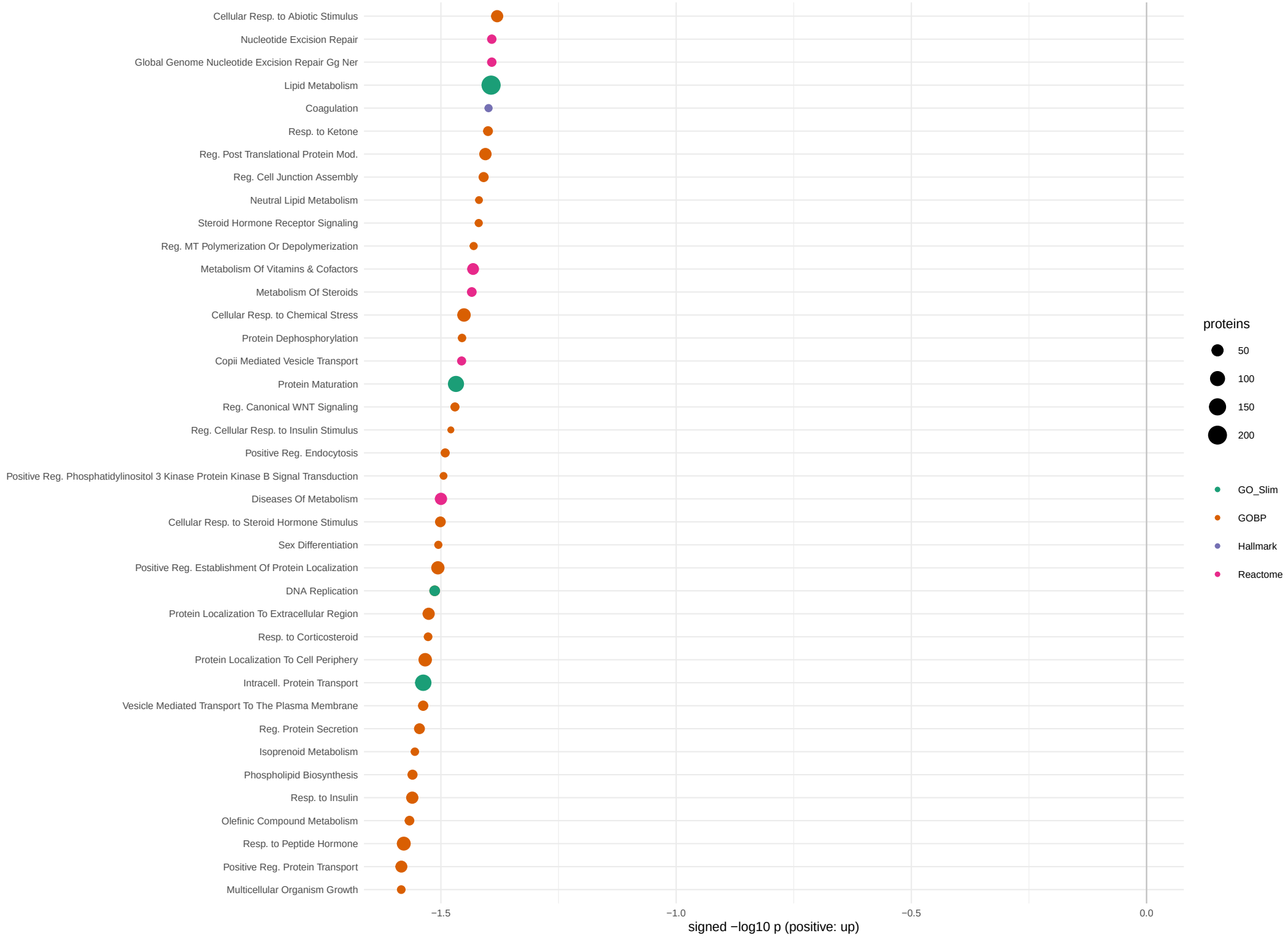

Acute_HR, fry sets at p < 0.05
233 sets. Page 5 of 6.
Positive Reg. Proteolysis
MT Polymerization Or Depolymerization
Positive Reg. Post Translational Protein Mod.
Positive Reg. Protein Containing Complex Assembly
Signaling By Robo Receptors
Reg. Cardiac Muscle Contraction By Calcium Ion Signaling
Positive Reg. Proteolysis Involved In Protein Catabolism
Ribosome Associated Quality Control
Reg. Release Of Sequestered Calcium Ion Into Cytosol By Sarcoplasmic Reticulum
UPR
Reg. Release Of Sequestered Calcium Ion Into Cytosol
Cellular Resp. to Heat
Negative Reg. RNA Catabolism
Cytoplasmic Translation
Reg. Apoptotic Signaling
Reg. Cell Projection Assembly
Reg. Cytoplasmic Translation
Positive Reg. Apoptotic Signaling
Resp. to Heat
Reg. MT Process
Sars Cov 1 Infection
Reg. Syncytium Formation By Plasma Membrane Fusion
Eukaryotic Translation Elongation
Reg. Ub−Dep. Protein Catabolism
Reg. Striated Muscle Contraction
Reg. Expression Of Slits & Robos
Translational Initiation
Reg. Proteas. Ub−Dep. Protein Catabolism
Reg. Protein Containing Complex Assembly
MT Polymerization
Reg. RNA Splicing
Embryonic Organ Development
Mitotic Spindle
Epigenetic Reg. Gene Expression
Cytoplasmic MT Org.
Negative Reg. mRNA Catabolism
Negative Reg. Ub−Dep. Protein Catabolism
Negative Reg. Syncytium Formation By Plasma Membrane Fusion
Nuclear Export
0.0
0.5
1.0
1.5
signed −log10 p (positive: up)
proteins
30
60
90
GO_Slim
GOBP
Hallmark
Reactome
fry on the imputed matrix with the fitted design, subject block and within−subject correlation; BH within contrast. Table: set_tests in c_data/set_tests.rds.


Acute_HR, fry sets at p < 0.05
233 sets. Page 6 of 6.
Viral Genome Replication
G2M Checkpoint
RNA Splicing Via Transesterification Reactions
mRNA Metabolism
MYC Targets V1
Reg. Viral Life Cycle
RNA Splicing
Dengue Virus Infection
Negative Reg. DNA Metabolism
Protein DNA Complex Org.
Nuclear Transcribed mRNA Catabolism
Negative Reg. Translation
Reg. mRNA Catabolism
Pre−mRNA Processing
Dengue Virus Host Interactions
Resp. to Virus
mRNA Splicing
Reg. mRNA Metabolism
Ribosome Quality Control Rqc Complex Extracts & Degrades Nascent Peptide
Reg. Viral Process
mRNA 3 End Processing
Translational Elongation
Membraneless Org. Assembly
mRNA Catabolism
Reg. The Force Of Heart Contraction
mRNA Polyadenylation
Resp. Syncytial Virus Infection Pathway
Negative Reg. mRNA Metabolism
Reg. Cardiac Muscle Contraction By Reg. The Release Of Sequestered Calcium Ion
Reg. Blood Circulation
Positive Reg. Protein Polymerization
Leukocyte Chemotaxis
Reg. Heart Contraction
0
1
2
3
signed −log10 p (positive: up)
proteins
30
60
90
120
GO_Slim
GOBP
Hallmark
Reactome
fry on the imputed matrix with the fitted design, subject block and within−subject correlation; BH within contrast. Table: set_tests in c_data/set_tests.rds.

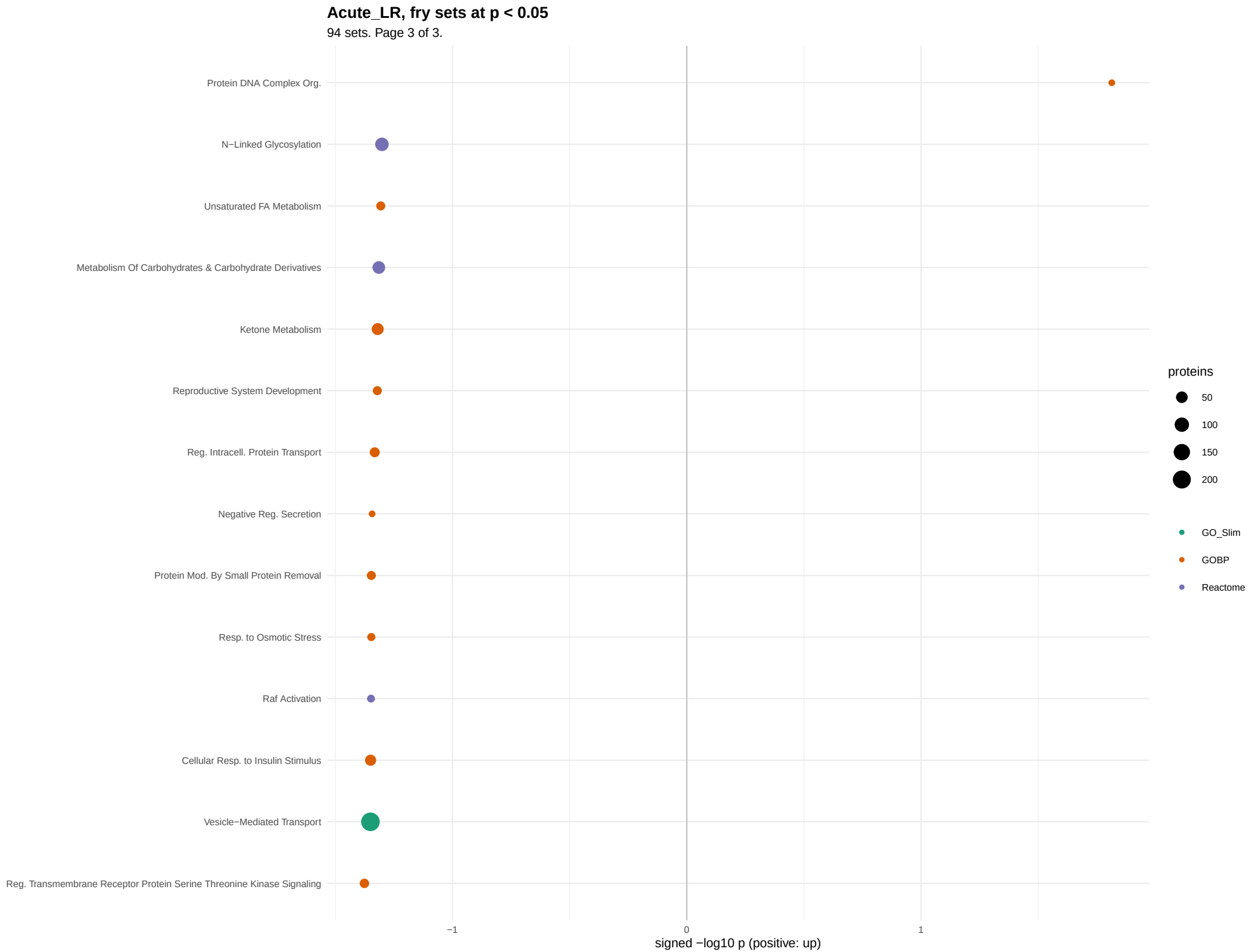

Acute_LR, fry sets at p < 0.05
94 sets. Page 1 of 3.
Adaptive Thermogenesis
Clathrin Endocytosis
DNA Damage Recognition In Gg Ner
Myd88 Independent Tlr4 Cascade
Pathways In Cancer
Glycerophospholipid Metabolism
Cytosolic Transport
Toll Like Receptor Cascades
Transforming Growth Factor Beta Receptor Signaling
Establishment Of Protein Localization To Membrane
Reg. Resp. to Cytokine Stimulus
Reg. Nervous System Development
Resp. to Transforming Growth Factor Beta
Retrograde Transport Endosome To Golgi
Phospholipid Metabolism
Resp. to Organophosphorus
Reg. MT Polymerization
Mitotic Spindle Checkpoint
Reg. Autophagy
Positive Reg. WNT Signaling
Endosomal Transport
Establishment Of PM Protein Localization
Sars Cov 2 Activates Modulates Innate & Adaptive Immune Responses
Endocytic Recycling
Glycerolipid Metabolism
Cell Surface Receptor Protein Serine Threonine Kinase Signaling
Hormone Mediated Signaling
Immune Response Regulating Cell Surface Receptor Signaling
Reg. Protein Transport
Ras Protein Signal Transduction
Reg. Neurogenesis
Temperature Homeostasis
Dephosphorylation
Signaling By Ntrks
Renal Cell Carcinoma
Protein Localization To Cell Junction
Protein Targeting
Gland Development
Reg. Lipid Localization
Resp. to Steroid Hormone
−2
−1
0
signed −log10 p (positive: up)
proteins
20
40
60
GOBP
KEGG_Legacy
Reactome
fry on the imputed matrix with the fitted design, subject block and within−subject correlation; BH within contrast. Table: set_tests in c_data/set_tests.rds.

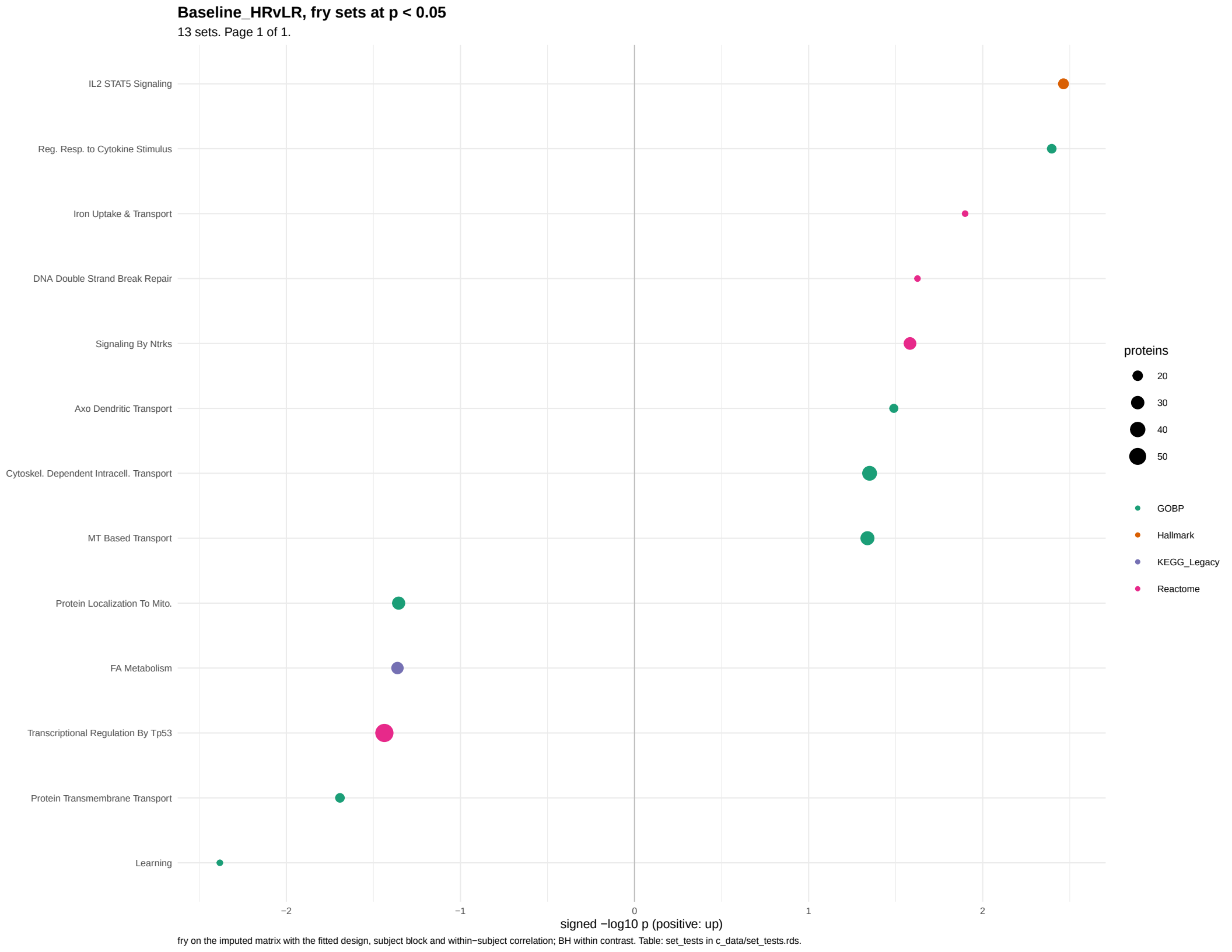
Acute_LR, fry sets at p < 0.05

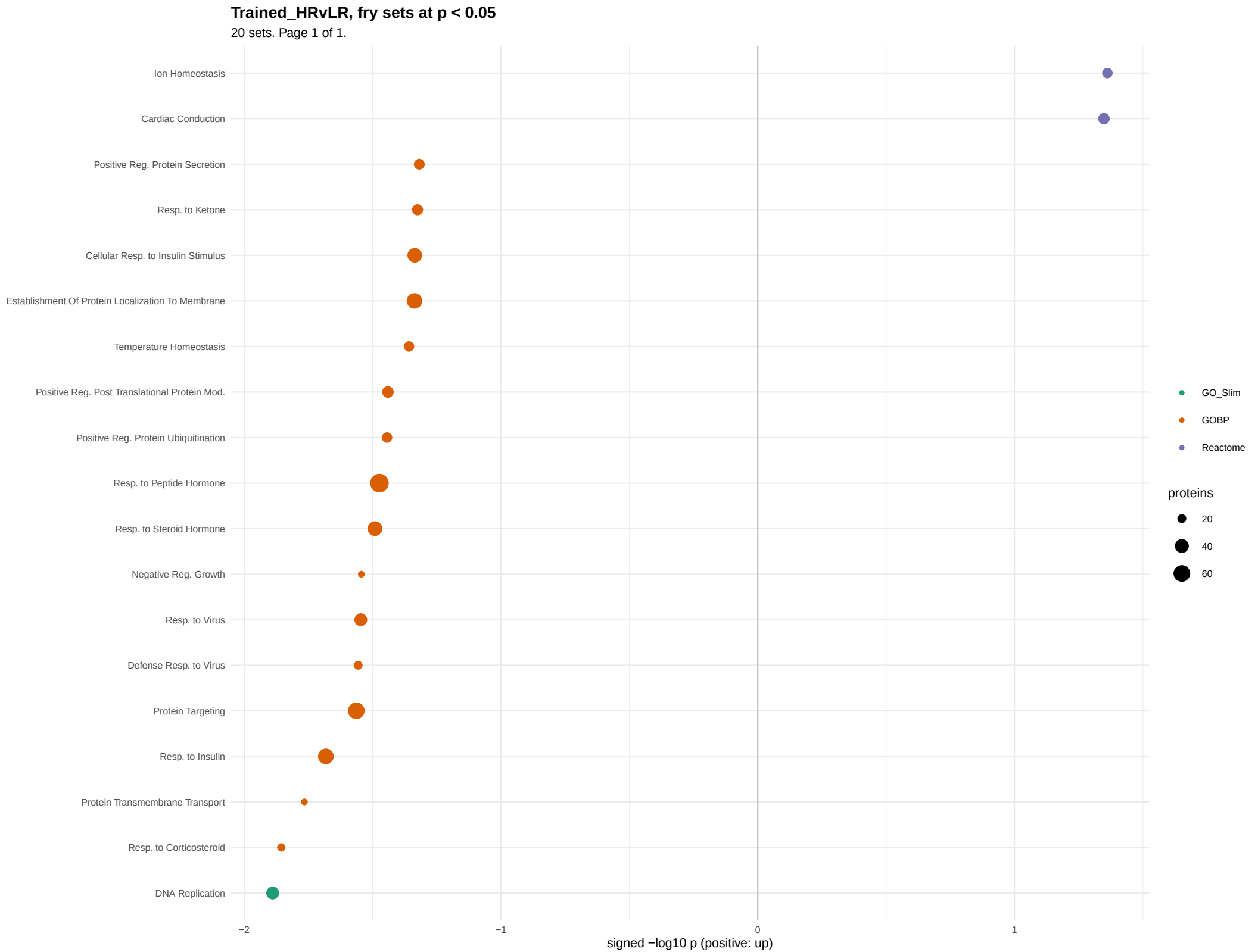
94 sets. Page 2 of 3.

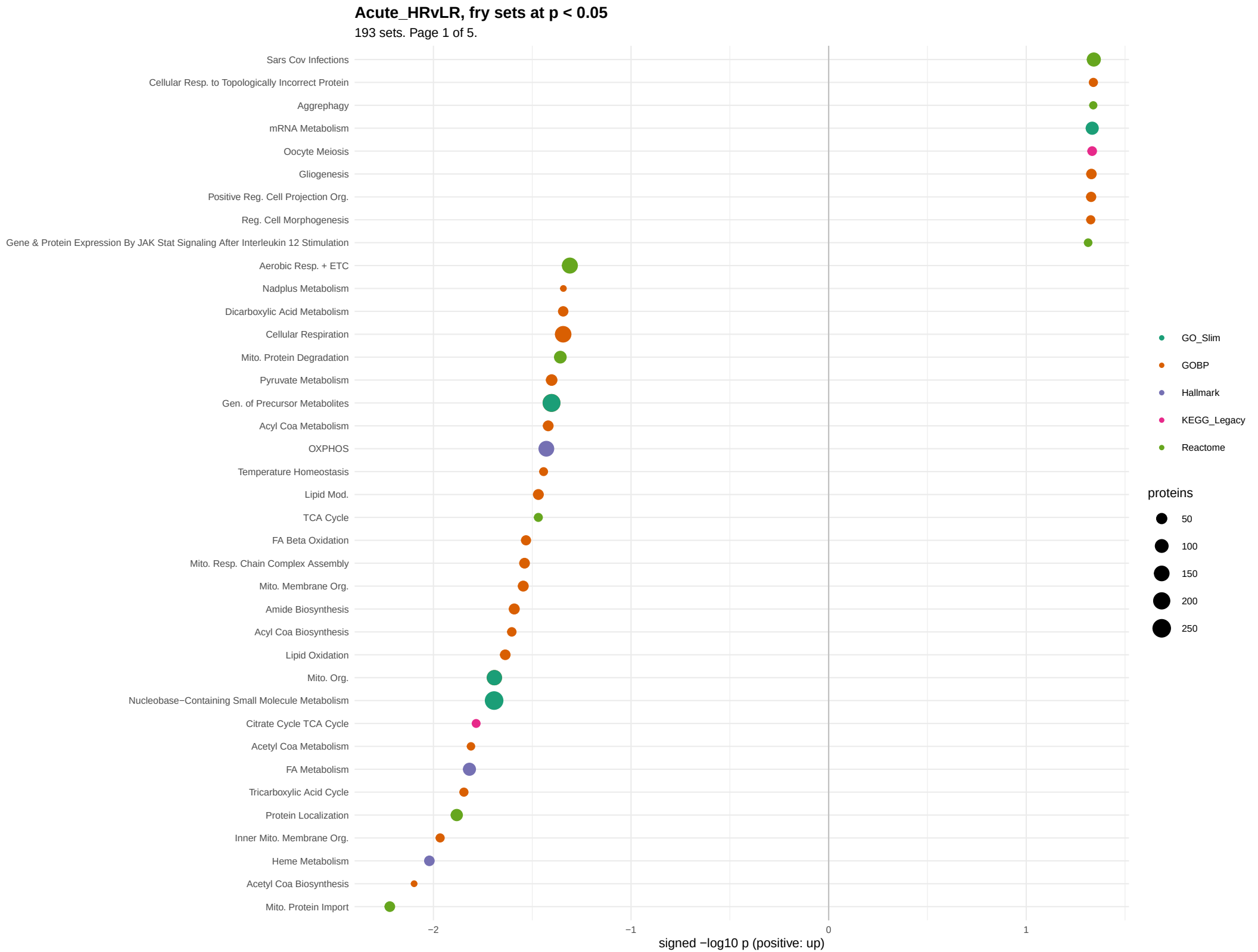


fry on the imputed matrix with the fitted design, subject block and within−subject correlation; BH within contrast. Table: set_tests in c_data/set_tests.rds.


Acute_LR, fry sets at p < 0.05
94 sets. Page 3 of 3.
Protein DNA Complex Org.
N−Linked Glycosylation
Unsaturated FA Metabolism
Metabolism Of Carbohydrates & Carbohydrate Derivatives
Ketone Metabolism
Reproductive System Development
Reg. Intracell. Protein Transport
Negative Reg. Secretion
Protein Mod. By Small Protein Removal
Resp. to Osmotic Stress
Raf Activation
Cellular Resp. to Insulin Stimulus
Vesicle−Mediated Transport
Reg. Transmembrane Receptor Protein Serine Threonine Kinase Signaling
−1
0
1
signed −log10 p (positive: up)
proteins
50
100
150
200
GO_Slim
GOBP
Reactome
fry on the imputed matrix with the fitted design, subject block and within−subject correlation; BH within contrast. Table: set_tests in c_data/set_tests.rds.

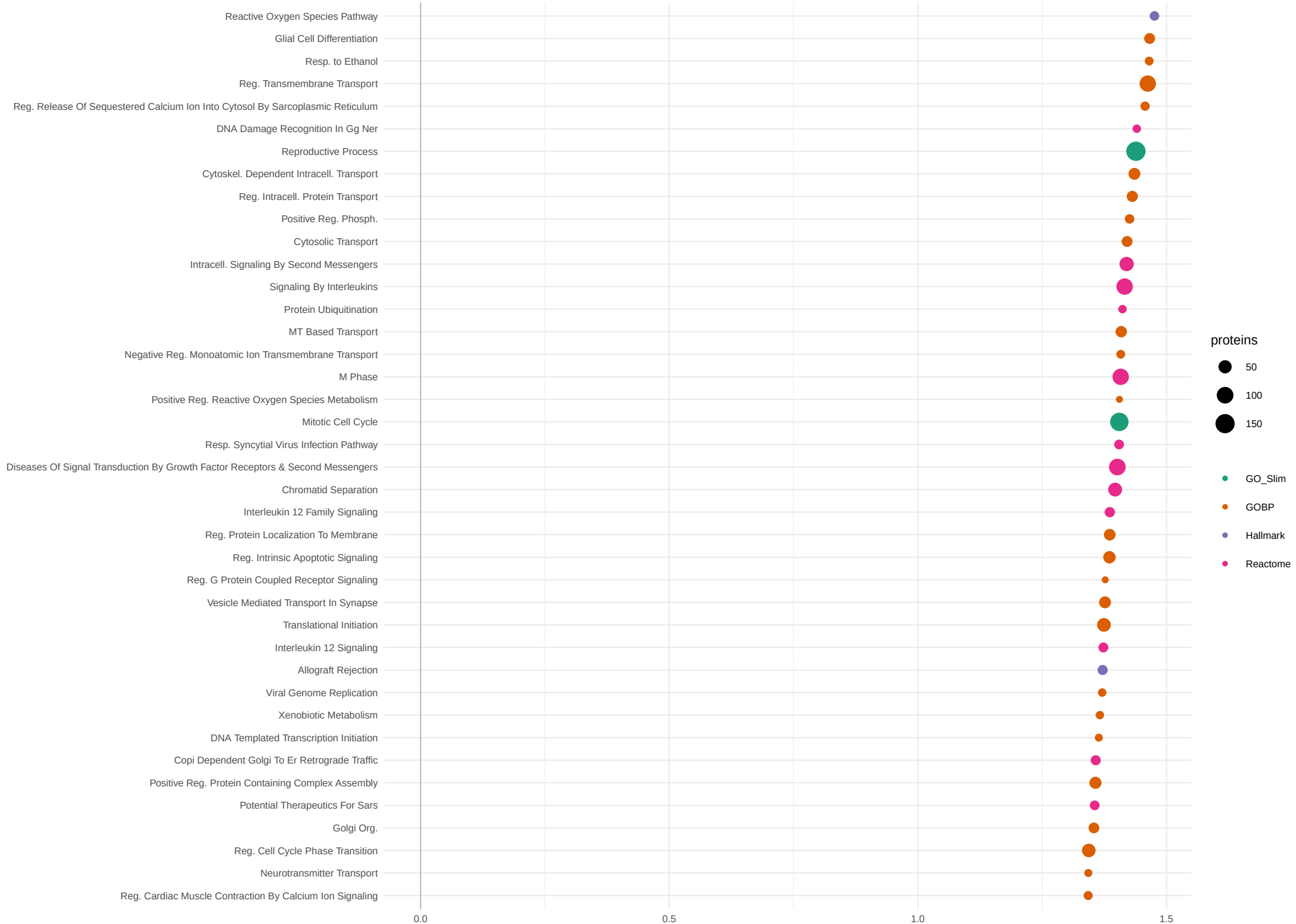

Baseline_HRvLR, fry sets at p < 0.05
13 sets. Page 1 of 1.
IL2 STAT5 Signaling
Reg. Resp. to Cytokine Stimulus
Iron Uptake & Transport
DNA Double Strand Break Repair
Signaling By Ntrks
Axo Dendritic Transport
Cytoskel. Dependent Intracell. Transport
MT Based Transport
Protein Localization To Mito.
FA Metabolism
Transcriptional Regulation By Tp53
Protein Transmembrane Transport
Learning
−2
−1
0
1
2
signed −log10 p (positive: up)
proteins
20
30
40
50
GOBP
Hallmark
KEGG_Legacy
Reactome
fry on the imputed matrix with the fitted design, subject block and within−subject correlation; BH within contrast. Table: set_tests in c_data/set_tests.rds.


Trained_HRvLR, fry sets at p < 0.05
20 sets. Page 1 of 1.
Ion Homeostasis
Cardiac Conduction
Positive Reg. Protein Secretion
Resp. to Ketone
Cellular Resp. to Insulin Stimulus
Establishment Of Protein Localization To Membrane
Temperature Homeostasis
Positive Reg. Post Translational Protein Mod.
Positive Reg. Protein Ubiquitination
Resp. to Peptide Hormone
Resp. to Steroid Hormone
Negative Reg. Growth
Resp. to Virus
Defense Resp. to Virus
Protein Targeting
Resp. to Insulin
Protein Transmembrane Transport
Resp. to Corticosteroid
DNA Replication
−2
−1
0
1
signed −log10 p (positive: up)
GO_Slim
GOBP
Reactome
proteins
20
40
60
fry on the imputed matrix with the fitted design, subject block and within−subject correlation; BH within contrast. Table: set_tests in c_data/set_tests.rds.

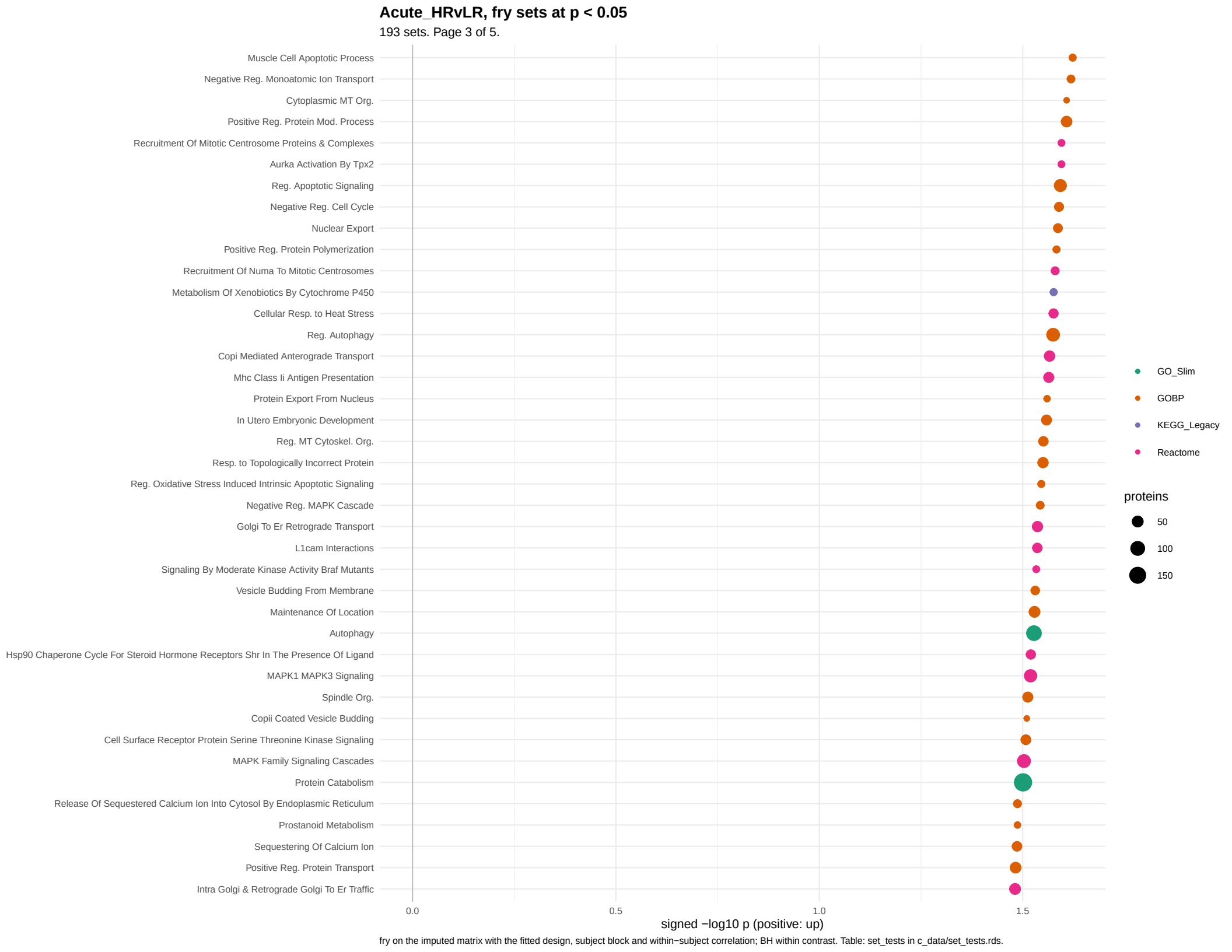

Acute_HRvLR, fry sets at p < 0.05
193 sets. Page 1 of 5.
Sars Cov Infections
Cellular Resp. to Topologically Incorrect Protein
Aggrephagy
mRNA Metabolism
Oocyte Meiosis
Gliogenesis
Positive Reg. Cell Projection Org.
Reg. Cell Morphogenesis
Gene & Protein Expression By JAK Stat Signaling After Interleukin 12 Stimulation
Aerobic Resp. + ETC
Nadplus Metabolism
Dicarboxylic Acid Metabolism
Cellular Respiration
Mito. Protein Degradation
Pyruvate Metabolism
Gen. of Precursor Metabolites
Acyl Coa Metabolism
OXPHOS
Temperature Homeostasis
Lipid Mod.
TCA Cycle
FA Beta Oxidation
Mito. Resp. Chain Complex Assembly
Mito. Membrane Org.
Amide Biosynthesis
Acyl Coa Biosynthesis
Lipid Oxidation
Mito. Org.
Nucleobase−Containing Small Molecule Metabolism
Citrate Cycle TCA Cycle
Acetyl Coa Metabolism
FA Metabolism
Tricarboxylic Acid Cycle
Protein Localization
Inner Mito. Membrane Org.
Heme Metabolism
Acetyl Coa Biosynthesis
Mito. Protein Import
−2
−1
0
1
signed −log10 p (positive: up)
GO_Slim
GOBP
Hallmark
KEGG_Legacy
Reactome
proteins
50
100
150
200
250
fry on the imputed matrix with the fitted design, subject block and within−subject correlation; BH within contrast. Table: set_tests in c_data/set_tests.rds.

Acute_HRvLR, fry sets at p < 0.05

193 sets. Page 2 of 5.

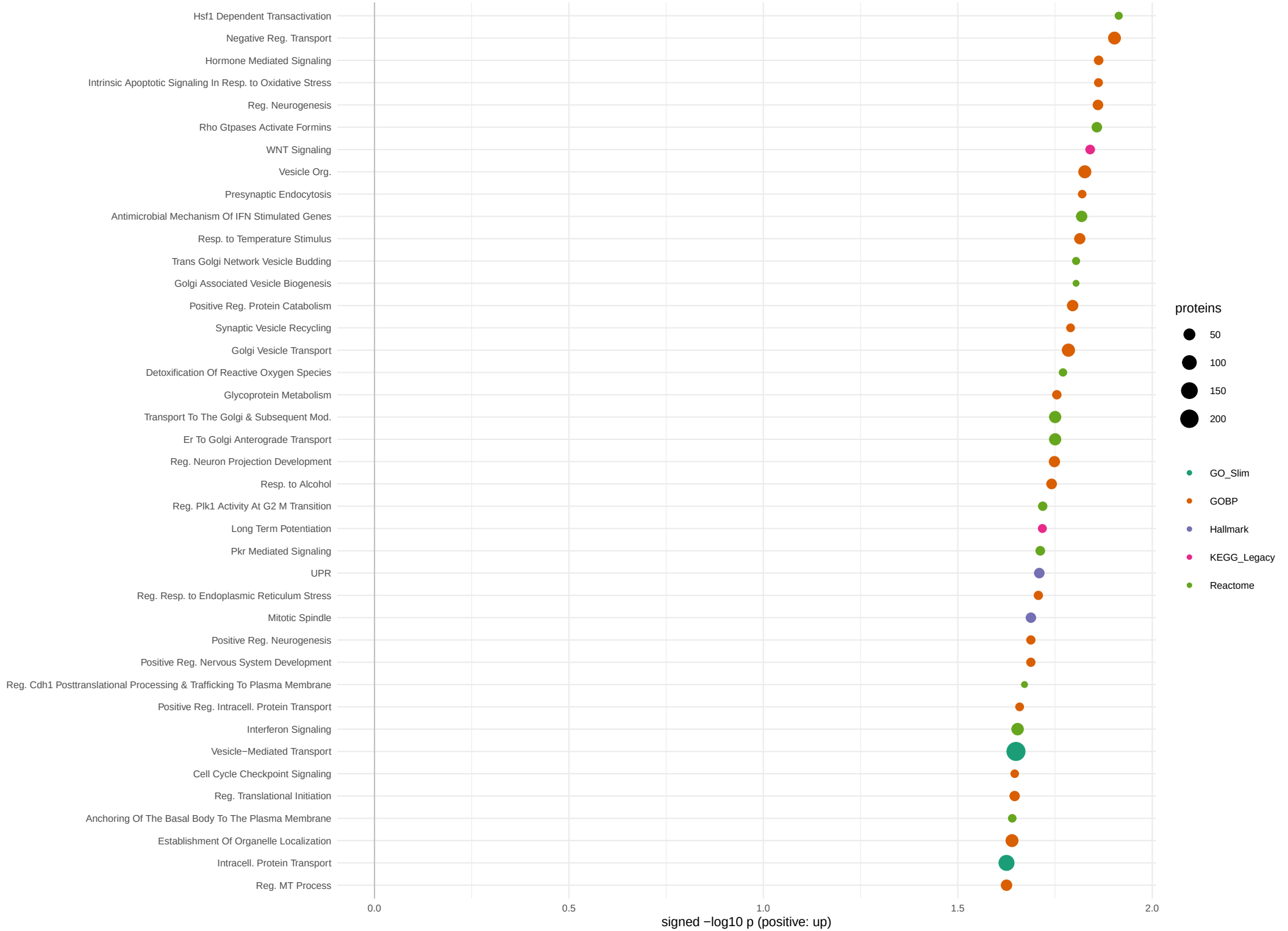


fry on the imputed matrix with the fitted design, subject block and within−subject correlation; BH within contrast. Table: set_tests in c_data/set_tests.rds.

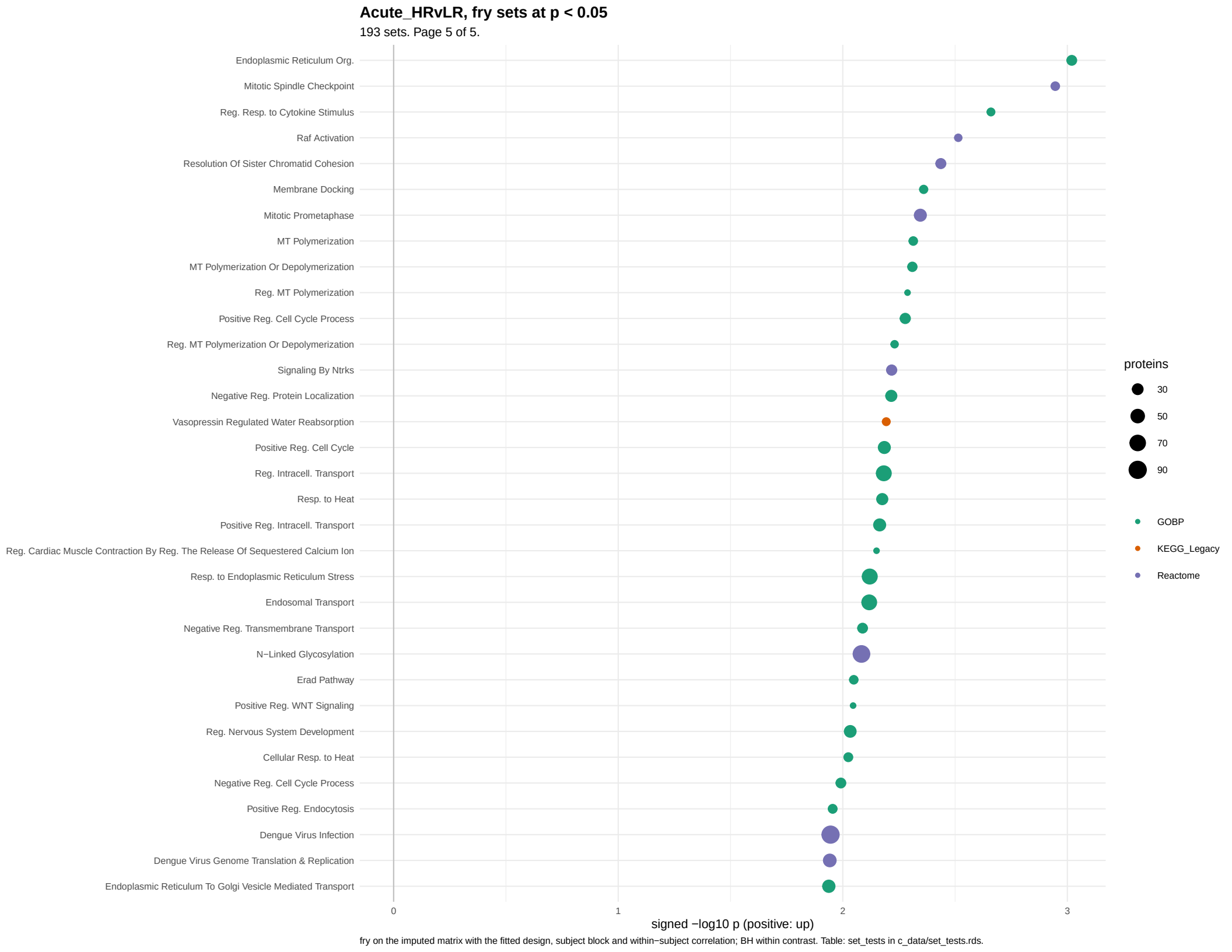

Acute_HRvLR, fry sets at p < 0.05
193 sets. Page 3 of 5.
Muscle Cell Apoptotic Process
Negative Reg. Monoatomic Ion Transport
Cytoplasmic MT Org.
Positive Reg. Protein Mod. Process
Recruitment Of Mitotic Centrosome Proteins & Complexes
Aurka Activation By Tpx2
Reg. Apoptotic Signaling
Negative Reg. Cell Cycle
Nuclear Export
Positive Reg. Protein Polymerization
Recruitment Of Numa To Mitotic Centrosomes
Metabolism Of Xenobiotics By Cytochrome P450
Cellular Resp. to Heat Stress
Reg. Autophagy
Copi Mediated Anterograde Transport
Mhc Class Ii Antigen Presentation
Protein Export From Nucleus
In Utero Embryonic Development
Reg. MT Cytoskel. Org.
Resp. to Topologically Incorrect Protein
Reg. Oxidative Stress Induced Intrinsic Apoptotic Signaling
Negative Reg. MAPK Cascade
Golgi To Er Retrograde Transport
L1cam Interactions
Signaling By Moderate Kinase Activity Braf Mutants
Vesicle Budding From Membrane
Maintenance Of Location
Autophagy
Hsp90 Chaperone Cycle For Steroid Hormone Receptors Shr In The Presence Of Ligand
MAPK1 MAPK3 Signaling
Spindle Org.
Copii Coated Vesicle Budding
Cell Surface Receptor Protein Serine Threonine Kinase Signaling
MAPK Family Signaling Cascades
Protein Catabolism
Release Of Sequestered Calcium Ion Into Cytosol By Endoplasmic Reticulum
Prostanoid Metabolism
Sequestering Of Calcium Ion
Positive Reg. Protein Transport
Intra Golgi & Retrograde Golgi To Er Traffic
0.0
0.5
1.0
1.5
signed −log10 p (positive: up)
GO_Slim
GOBP
KEGG_Legacy
Reactome
proteins
50
100
150
fry on the imputed matrix with the fitted design, subject block and within−subject correlation; BH within contrast. Table: set_tests in c_data/set_tests.rds.

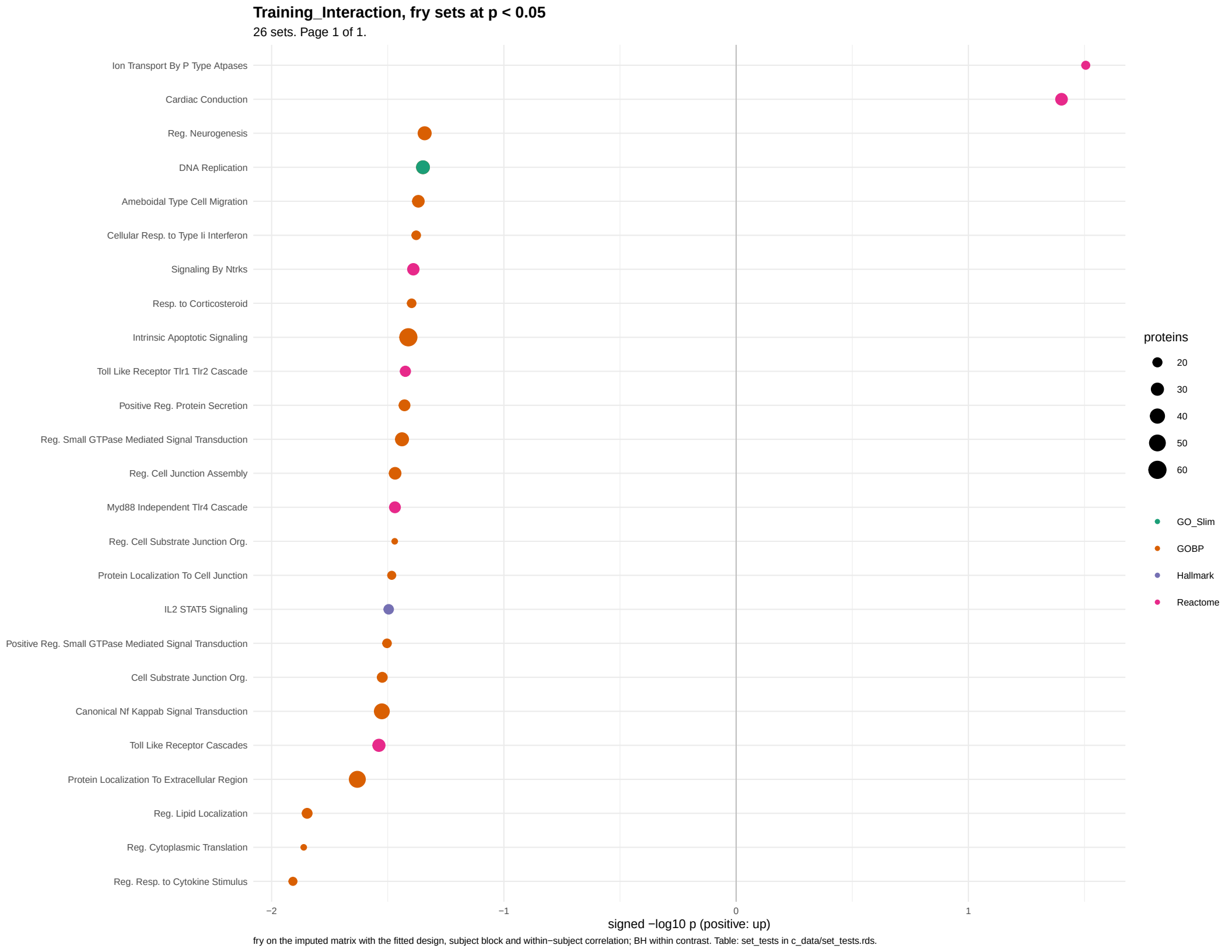
Acute_HRvLR, fry sets at p < 0.05

193 sets. Page 4 of 5.

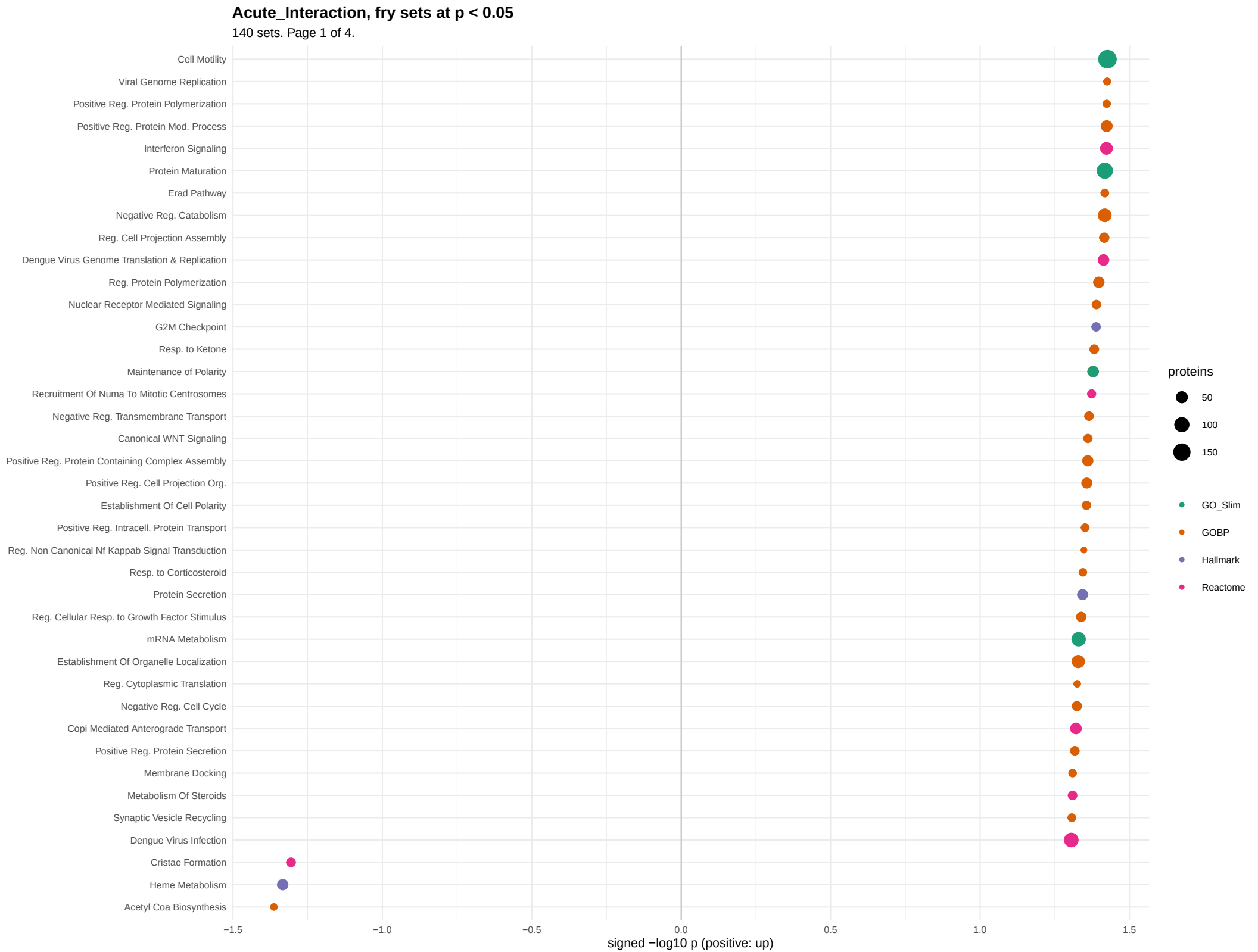


fry on the imputed matrix with the fitted design, subject block and within−subject correlation; BH within contrast. Table: set_tests in c_data/set_tests.rds.

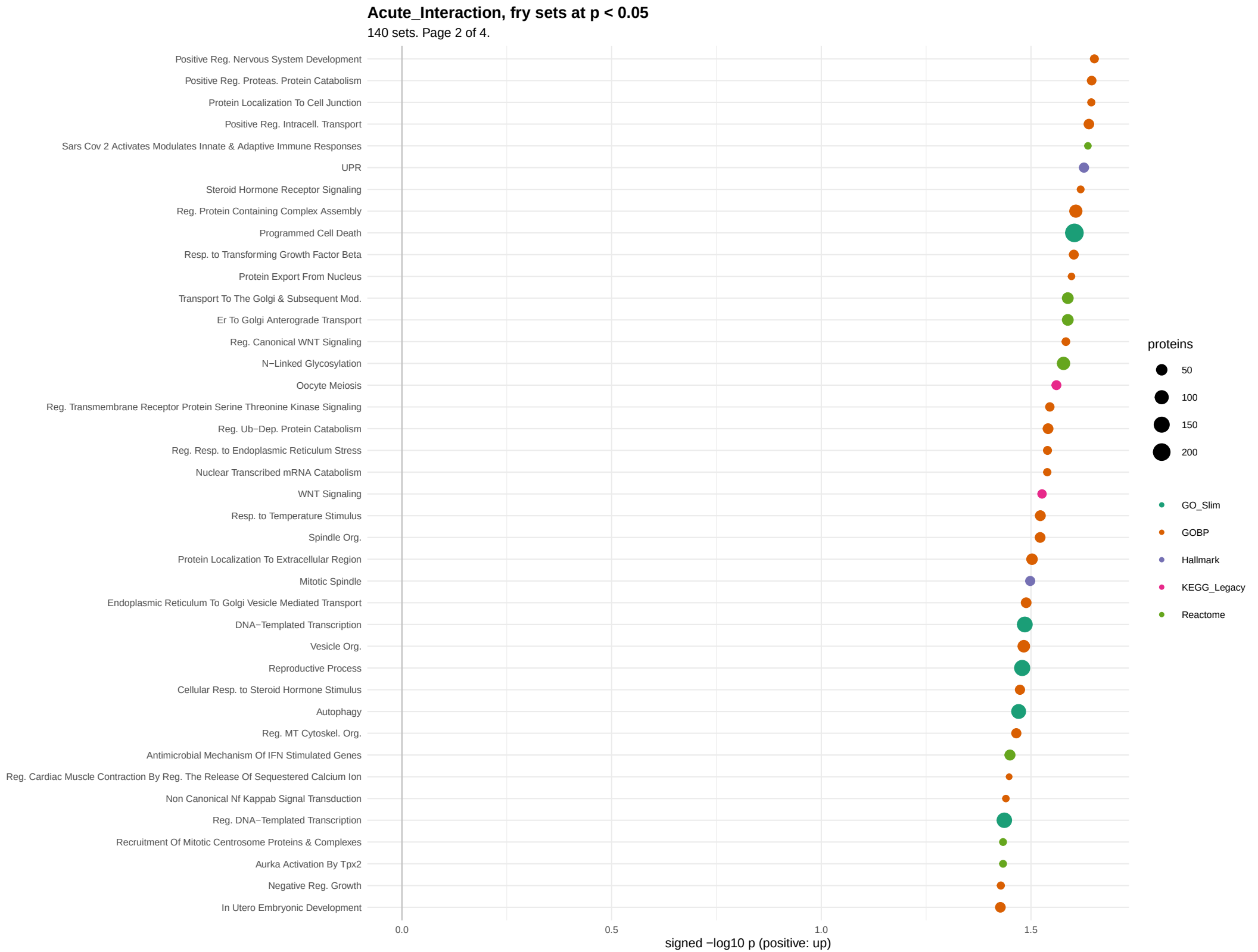

Acute_HRvLR, fry sets at p < 0.05
193 sets. Page 5 of 5.
Endoplasmic Reticulum Org.
Mitotic Spindle Checkpoint
Reg. Resp. to Cytokine Stimulus
Raf Activation
Resolution Of Sister Chromatid Cohesion
Membrane Docking
Mitotic Prometaphase
MT Polymerization
MT Polymerization Or Depolymerization
Reg. MT Polymerization
Positive Reg. Cell Cycle Process
Reg. MT Polymerization Or Depolymerization
Signaling By Ntrks
Negative Reg. Protein Localization
Vasopressin Regulated Water Reabsorption
Positive Reg. Cell Cycle
Reg. Intracell. Transport
Resp. to Heat
Positive Reg. Intracell. Transport
Reg. Cardiac Muscle Contraction By Reg. The Release Of Sequestered Calcium Ion
Resp. to Endoplasmic Reticulum Stress
Endosomal Transport
Negative Reg. Transmembrane Transport
N−Linked Glycosylation
Erad Pathway
Positive Reg. WNT Signaling
Reg. Nervous System Development
Cellular Resp. to Heat
Negative Reg. Cell Cycle Process
Positive Reg. Endocytosis
Dengue Virus Infection
Dengue Virus Genome Translation & Replication
Endoplasmic Reticulum To Golgi Vesicle Mediated Transport
0
1
2
3
signed −log10 p (positive: up)
proteins
30
50
70
90
GOBP
KEGG_Legacy
Reactome
fry on the imputed matrix with the fitted design, subject block and within−subject correlation; BH within contrast. Table: set_tests in c_data/set_tests.rds.

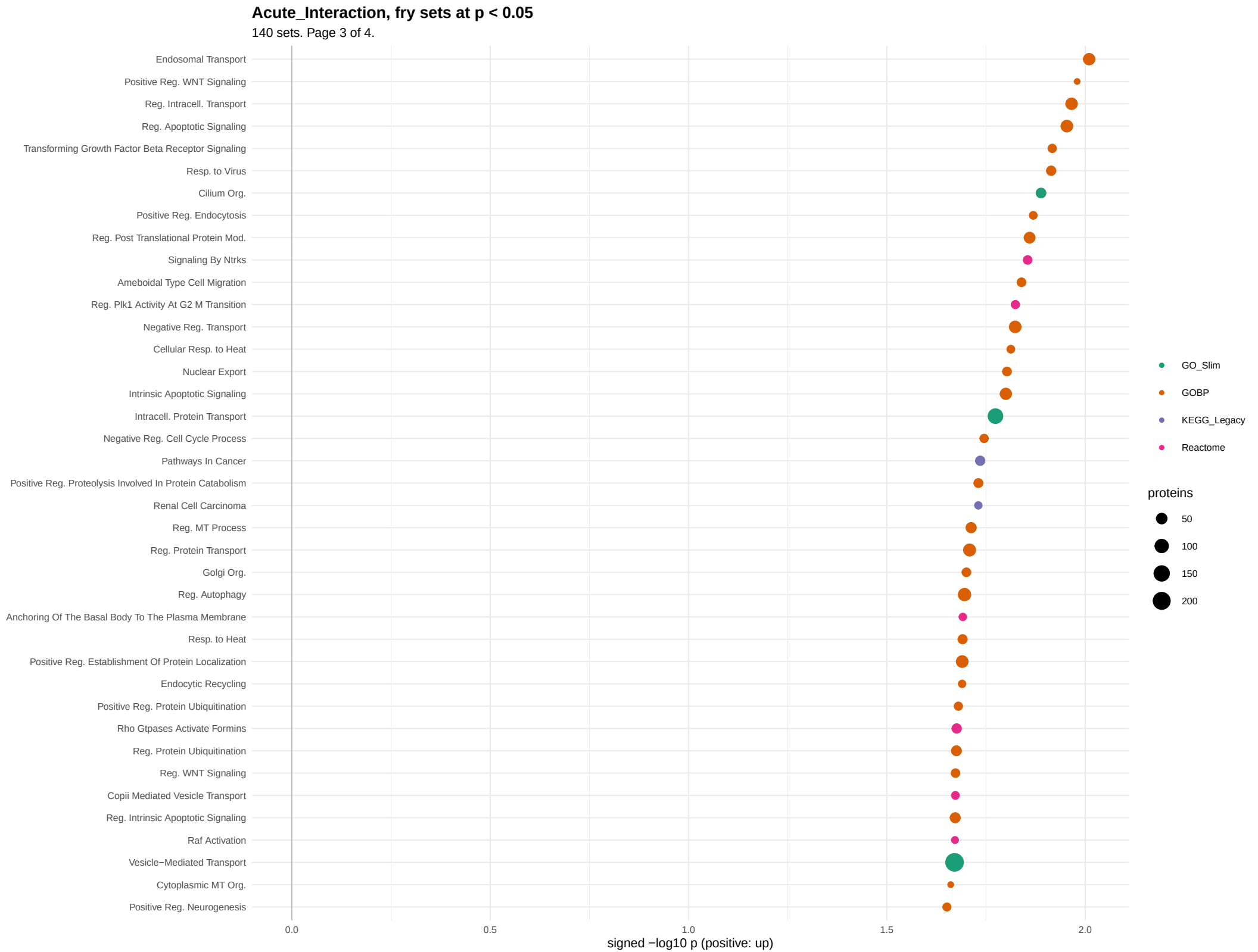


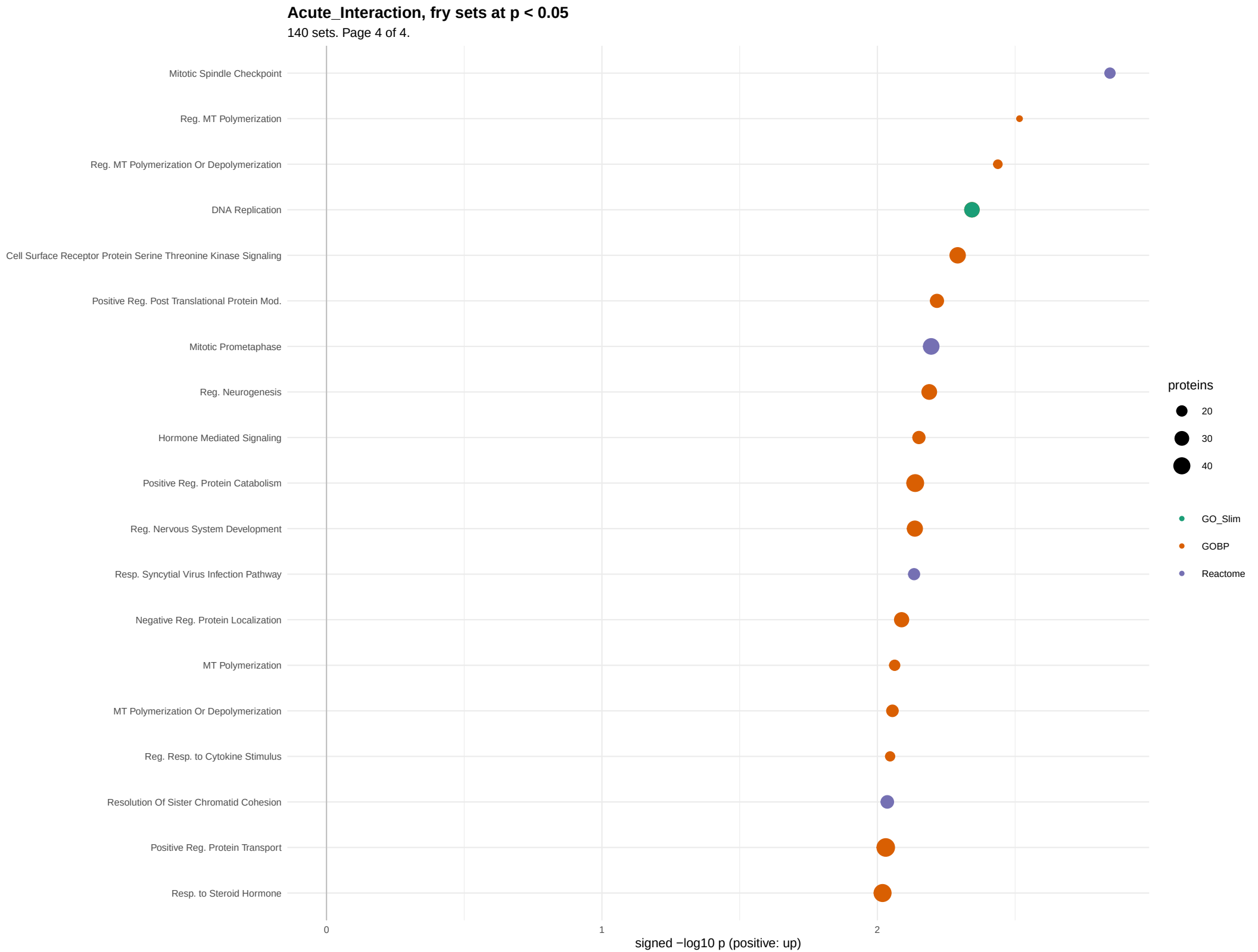
fry on the imputed matrix with the fitted design, subject block and within−subject correlation; BH within contrast. Table: set_tests in c_data/set_tests.rds.



fry on the imputed matrix with the fitted design, subject block and within−subject correlation; BH within contrast. Table: set_tests in c_data/set_tests.rds.


Acute_Interaction, fry sets at p < 0.05
140 sets. Page 2 of 4.
Positive Reg. Nervous System Development
Positive Reg. Proteas. Protein Catabolism
Protein Localization To Cell Junction
Positive Reg. Intracell. Transport
Sars Cov 2 Activates Modulates Innate & Adaptive Immune Responses
UPR
Steroid Hormone Receptor Signaling
Reg. Protein Containing Complex Assembly
Programmed Cell Death
Resp. to Transforming Growth Factor Beta
Protein Export From Nucleus
Transport To The Golgi & Subsequent Mod.
Er To Golgi Anterograde Transport
Reg. Canonical WNT Signaling
N−Linked Glycosylation
Oocyte Meiosis
Reg. Transmembrane Receptor Protein Serine Threonine Kinase Signaling
Reg. Ub−Dep. Protein Catabolism
Reg. Resp. to Endoplasmic Reticulum Stress
Nuclear Transcribed mRNA Catabolism
WNT Signaling
Resp. to Temperature Stimulus
Spindle Org.
Protein Localization To Extracellular Region
Mitotic Spindle
Endoplasmic Reticulum To Golgi Vesicle Mediated Transport
DNA−Templated Transcription
Vesicle Org.
Reproductive Process
Cellular Resp. to Steroid Hormone Stimulus
Autophagy
Reg. MT Cytoskel. Org.
Antimicrobial Mechanism Of IFN Stimulated Genes
Reg. Cardiac Muscle Contraction By Reg. The Release Of Sequestered Calcium Ion
Non Canonical Nf Kappab Signal Transduction
Reg. DNA−Templated Transcription
Recruitment Of Mitotic Centrosome Proteins & Complexes
Aurka Activation By Tpx2
Negative Reg. Growth
In Utero Embryonic Development
0.0
0.5
1.0
1.5
signed −log10 p (positive: up)
proteins
50
100
150
200
GO_Slim
GOBP
Hallmark
KEGG_Legacy
Reactome
fry on the imputed matrix with the fitted design, subject block and within−subject correlation; BH within contrast. Table: set_tests in c_data/set_tests.rds.

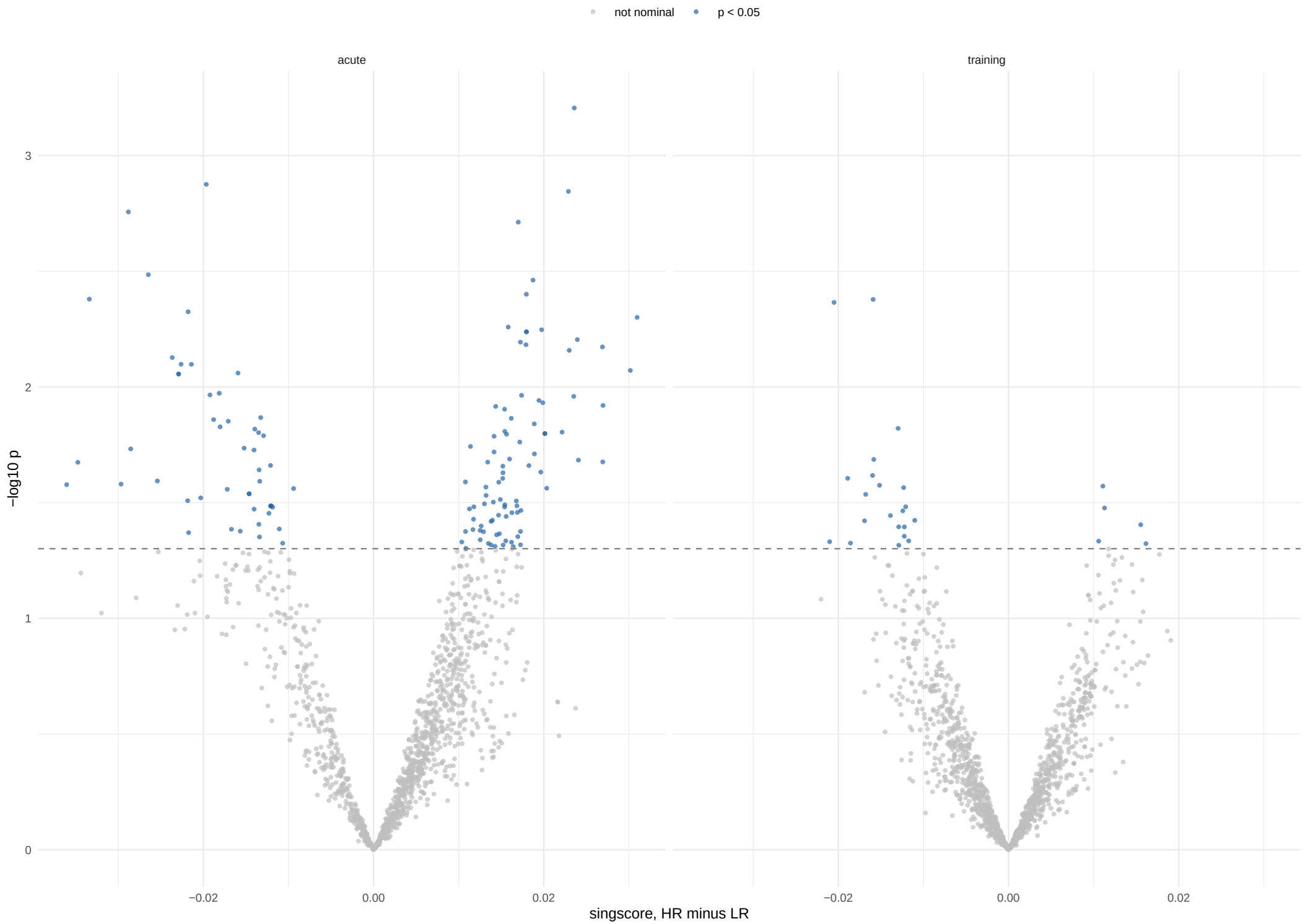

Acute_Interaction, fry sets at p < 0.05
140 sets. Page 3 of 4.
Endosomal Transport
Positive Reg. WNT Signaling
Reg. Intracell. Transport
Reg. Apoptotic Signaling
Transforming Growth Factor Beta Receptor Signaling
Resp. to Virus
Cilium Org.
Positive Reg. Endocytosis
Reg. Post Translational Protein Mod.
Signaling By Ntrks
Ameboidal Type Cell Migration
Reg. Plk1 Activity At G2 M Transition
Negative Reg. Transport
Cellular Resp. to Heat
Nuclear Export
Intrinsic Apoptotic Signaling
Intracell. Protein Transport
Negative Reg. Cell Cycle Process
Pathways In Cancer
Positive Reg. Proteolysis Involved In Protein Catabolism
Renal Cell Carcinoma
Reg. MT Process
Reg. Protein Transport
Golgi Org.
Reg. Autophagy
Anchoring Of The Basal Body To The Plasma Membrane
Resp. to Heat
Positive Reg. Establishment Of Protein Localization
Endocytic Recycling
Positive Reg. Protein Ubiquitination
Rho Gtpases Activate Formins
Reg. Protein Ubiquitination
Reg. WNT Signaling
Copii Mediated Vesicle Transport
Reg. Intrinsic Apoptotic Signaling
Raf Activation
Vesicle−Mediated Transport
Cytoplasmic MT Org.
Positive Reg. Neurogenesis
0.0
0.5
1.0
1.5
2.0
signed −log10 p (positive: up)
GO_Slim
GOBP
KEGG_Legacy
Reactome
proteins
50
100
150
200
fry on the imputed matrix with the fitted design, subject block and within−subject correlation; BH within contrast. Table: set_tests in c_data/set_tests.rds.



fry on the imputed matrix with the fitted design, subject block and within−subject correlation; BH within contrast. Table: set_tests in c_data/set_tests.rds.

ANCOVA: HR minus LR at the later time point, at equal starting value

Training: 26 of 1378 sets at p < 0.05. Acute: 135 of 1378.



lm(later ~ earlier + arm) per set; training is T2 on T1, acute T3 on T2. Dashed line: p = 0.05. BH within window and collection. Table: c_data/02_contrasts.xlsx, ancova.